

American International University-Bangladesh (AIUB)

Faculty of Science and Technology

Beyond Hand Gestures: A Hybrid Deep Learning Framework for Joint Sign Language and Facial Emotion Recognition

Md Shahariar Kabir Bhuiya (22-49513-3)

Jiad Hasan(22-48563-3)

Faisal Ahmed (22-49591-3)

Md Abdul Alim (23-51105-1)

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Declaration

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Md Shahariar Kabir Bhuiya
22-49513-3
Department of CSE, AIUB

Jiad Hasan
22-48563-3
Department of CSE, AIUB

Faisal Ahmed
22-49591-3
Department of CSE, AIUB

Md. Abdul Alim
23-51105-1
Department of CSE, AIUB

Approval

The thesis titled “Beyond Hand Gestures: A Hybrid Deep Learning Framework for Joint Sign Language and Facial Emotion Recognition” has been submitted to the following respected members of the board of examiners of the department of computer science in partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science on October, 2026 and has been accepted as satisfactory.

Md. Hamid Uddin

Lecturer & Supervisor
Department of Computer Science
American International University-Bangladesh

Md.Ariful Islam

Lecturer & External
Department of Computer Science
American International University-Bangladesh

Prof. Dr. Muhammad Firoz Mridha
Professor & Head (UG)
Department of Computer Science
American International University-Bangladesh

Prof. Dr. Khandaker Tabin Hasan
Professor & Associate Dean
Faculty of Science and Technology
American International University-Bangladesh

Prof. Dr. Dip Nandi
Professor & Dean
Faculty of Science and Technology
American International University-Bangladesh

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List of Abbreviations

Abbreviation / Symbol	Full Form / Technical Description
AIUB	American International University-Bangladesh
ASL	American Sign Language
ArSL	Arabic Sign Language
BdSL	Bangla Sign Language
Bi-LSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
CSL	Chinese Sign Language
CSLT	Continuous Sign Language Translation
CTC	Connectionist Temporal Classification
CUDA	Compute Unified Device Architecture
FER	Facial Emotion Recognition
FN / FP	False Negative / False Positive
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GNN	Graph Neural Network
GPU	Graphics Processing Unit
GRU	Gated Recurrent Unit
HCI	Human-Computer Interaction
HMM	Hidden Markov Model
HOG	Histogram of Oriented Gradients
IMU	Inertial Measurement Unit
ISL	Indian Sign Language
LSE	Spanish Sign Language (Lengua de Signos Española)
LSTM	Long Short-Term Memory
mAP	Mean Average Precision
MLP	Multilayer Perceptron
MSA	Multi-Head Self-Attention
MTDLA	Multi-Task Deep Learning Architecture
RGB	Red, Green, and Blue Color Channels
ROI	Region of Interest
SIFT	Scale-Invariant Feature Transform
SLR	Sign Language Recognition
ST-GCN	Spatial-Temporal Graph Convolutional Network
SVM	Support Vector Machine
TCN	Temporal Convolutional Network
TP / TN	True Positive / True Negative

ViT	Vision Transformer
WBS	Work Breakdown Structure

Abstract

- Sign language is a crucial visual-gestural language used by millions of Deaf and Hard of Hearing people all over the world, which is a mixture of manual hand gestures and non-manual facial expressions. In this thesis two related but separately implemented tasks are explored: Dynamic sign-language recognition for 47 sign classes and Facial emotion recognition for 3 emotions (angry, neutral and sad). The lightweight facial emotion model (0.91M parameters) only employs a GNN and TCN without the Vision Transformer (ViT), which is used to extract the 3D facial landmark sequences from MediaPipe.
- To achieve this, our methodology contains four core computational stages in two task-specific pipelines – (1) video preprocessing and three-dimensional anatomical landmark extraction using Google MediaPipe on hands (21 keypoints per hand), facial structures (478 anatomical landmarks) and body pose (33 anchors); (2) patch-level global visual representation learning using Multi-Head Self-Attention in a Vision Transformer (ViT) in the sign-language branch; (3) spatial-topological relationship modeling across anatomical joints and facial muscle regions using a Graph Neural Network (GNN); and (4) long-range sequential kinematic modeling using dilated causal one-dimensional convolutions in a Temporal Convolutional Network (TCN) followed by task-specific Softmax classification heads.
- Separate quantitative evaluation was carried out for the two task-specific models. On dynamic sign-language recognition (30 training epochs, best validation accuracy 97.21% at Epoch 25), the trained checkpoint (best.pt) was tested on two different held-out partitions: 430-clip held-out split defined by split. json (96.74% test accuracy and 96.77% macro F1 score) and the separate 336-clip./data/test evaluation folder (96.43% test accuracy, 96.47% macro F1 score, 95.41% Softmax confidence on single-video inference ('same')). The 11-epoch GNN-TCN with 144 landmark clips (partitioned into 102 training, 21 validations, and 21 test clips) had an 80.95% accuracy for test clips (17/21 correct), a 100% validation accuracy, and a 96.08% training accuracy with 79.26% macro F1-score for facial emotion recognition. The emotion results must be interpreted with caution, since it is a within-subjects clip-level test with 21 clips and the dataset log is labeled for the subject ID. The Sign Dataset is split into two parts: (A) 430 clips from 47 classes in split. json and (B) 336 clips from 47 classes in ./data/test.

Keywords

sign language recognition, facial emotion recognition, vision transformer, graph neural network, temporal convolutional network, MediaPipe landmarks

Chapter 1: Introduction

1.1 Introduction

Human communication is a natural multi modal phenomenon, where words are continuously modulated by affective facial expression, body posture and temporal prosody. Sign language is the first language of over 70 million people worldwide who are deaf or hard of hearing, and hundreds of thousands who use sign language in Bangladesh. Sign languages have no phonetic or acoustic basis, unlike spoken languages, but are purely visual-spatial. In this visual modality, complete semantic meaning arises from the synchronously combined two channels: manual components (including hand shape, palm orientation, finger articulation, and spatial trajectory of the movement) and non-manual components (including facial expression, furrowed or raised eyebrows, eye aperture changes, morphemes of the mouth, head tilts, and upper body shifts).

Even though the visual-gestural languages like the American Sign Language (ASL), Bangla Sign Language (BdSL) and Indian Sign Language (ISL) are rich in grammar, there are profound communication barriers between deaf signers and the non-signing majority. The large majority of hearing citizens, medical practitioners, educators, and administrators are not fluent in sign language, which means that all too often, deaf individuals face systemic exclusion in their health care consultations, the classroom, in legal proceedings, and in the day-to-day civic life. Human sign language interpreters fulfil a vital link, but they are few and expensive, and are not readily available for everyday communication or for clinical triage in emergency situations.

In the context of computer vision, affective computing and assistive human-computer interaction (HCI), automated Sign Language Recognition (SLR) aims to fill this communication gap using artificial intelligence. But moving beyond individual hand postures, we need to see expressive human signing and translate it into correct interpretations of meaning. If a signer adopts a dynamic lexical gesture, the facial expression accompanying the gesture can indicate whether the statement is a calm declaration, an urgent question, a frustrated or angry expression or a sad expression. Thus, an intelligent computational framework that is able to interpret both dynamic manual sign language and non-manual facial expressions of emotions using dedicated task specific neural architectures is needed for the development of context-aware human-centric assistive communication.

Sign languages have underlying complex phonological and morphological aspects with suprasegmental prosodic markers in form of non-manual signals as well as lexical hand movements. It has been shown by linguistic analysis of both American Sign Language (ASL) and Bangla Sign Language (BdSL) that the same manual trajectory can have very different meaning in the context of different facial configuration. For example, raised eyebrows and forward head tilt generally indicate polar (yes/no) interrogatives, furrowed brow indicates wh-questions (who, what, where), and lowered lip corners and narrowed eyelids indicate sadness, physical pain, or emotional distress. Once these automated computer vision pipelines ignore the facial landmarks and only return a bounding box around the hand, this layer of pragmatic and emotional meaning is gone.

What is more, the socioeconomic factors of developing countries like Bangladesh makes accessible, camera-based automated sign interpretation a pressing need. Hundreds of thousands of hearing-impaired citizens use sign language as their primary mode of communication, but there are virtually no special interpretation services in public hospitals, in lecture halls at universities, at banking counters or at judicial courts. The development of automated computer vision systems capable of detecting manual and non-manual emotional states is thus an important issue for inclusive interaction between humans and computers.

1.2 Problem Statement

Existing automated sign language recognition systems focus on the classification of manual hand-gestures, which are highly context dependent, while largely ignoring the simultaneous spatial appearance, anatomical joint topology and long-range temporal continuity of the non-manual facial emotional state in the problem of sign language recognition. There are four technical shortcomings that are closely related and characterize this problem. On one hand, classical static image classifiers and frame-wise object detectors consider signing videos as uncorrelated 2D matrices and are easily affected by background clutter, illumination changes, and clothing changes of the signers, and they are not able to capture the sequential movement trajectories in which the dynamic multi-frame words are embedded. Second, current pose-based pipelines which extract skeletal coordinates tend to reduce 3D joint locations to unordered 1D vectors which cannot capture the intrinsic anatomical graph topology that guides the inter-joint articulations between fingers and the deformations of facial muscles. Third, ordinary recurrent sequence models have sequential computation bottlenecks and have a problem of gradient decay over long multi-frame signing windows. Fourth, existing vision-based sign recognition systems do not consider non-manual facial affect analysis at all and even the existing baseline systems which are limited to sparse

landmarks have difficulty in differentiating between emotionally neutral or negative-valence facial expressions, which leads to baseline or emotionally flat interpretation and therefore incomplete interpretation in the context of sign recognition.

At the basic level of computation, this problem is still lingering because in the case of manual gesture recognition and non-manual facial emotion recognition, there is a conflict in the representational demands that cannot be addressed by single paradigm models. Grid-based models are easily fooled by background clutter, and the coordinate sequence models lack any physical bone and muscle connectivity between the points representing finger joints and facial landmarks, whereas the static model is missing the temporal continuity between frames. Moreover, when facial expressions are only modeled by 3D sparse landmarks, changes in the eyebrows in the base expression (neutral) can be close to the geometric limit of changes in the furrowed-brow expression (anger). Automated sign language recognition is incomplete without a principled framework that takes into account the appearance of the sign language in the world, anatomical graph topology, parallelizable temporal dynamics and landmark-based facial affect modeling.

1.3 Problem Background

The automated sign language recognition systems based on technology can be historically traced along two main technological directions: one is wearable sensor-based systems and the other is camera-based computer vision systems. Wearable sensor systems use an electromechanical data-glove, inertial measurement units and flex transducers on the signers' hands to allow the direct measurement of the bending angles of the fingers and wrist rotation. While sensor gloves can be directly used to measure mechanics and are not affected by changes in the lighting, they have significant limitations in practical use as they are expensive to make, physically constraining and uncomfortable to wear, and they can lose their calibration over time. In addition, they cannot detect the facial expression or head movement that occur naturally.

Camera-based computer vision pipelines operate on regular RGB video feeds, removing the need for an intrusive wearable hardware. Stationary alphabet cards and digits have been classified in the earlier visual system using colour space skin segmentation and handcrafted geometric description. Although they can perform well with individual single-hand fingerspelling in a controlled environment with homogenous background, the static image classifiers are not suitable for the continuous and dynamic word-level signing, where the

meaning is conveyed through the direction of the movement, velocity of the hand and transitions between frames. Recurrent neural networks were added to the subsequent sequential pipelines to model temporal sequences from a set of extracted body keypoints. Recurrent architectures, however, do step-by-step processing of frames and suffer from high training latency and decay in memory for longer signing clips.

Moreover, within the framework of multimodal communication, many studies in linguistics and psychology have already shown that facial expression in sign languages have two functions: grammatical and emotional. However, previous multimodal methods have either been based on auxiliary vocal audio inputs that are not relevant in deaf communication, or on only visual processing of the hands. To gain a deeper understanding of this problem, let's trace the roots of sequence modeling through history. Recurrent neural networks (RNNs) in the form of Long Short-Term Memory (LSTMs) and Gated Recurrent Unit (GRUs) were used for several years as the standard way to model temporal video sequences. But recurrent networks are not parallelizable across time during training, and effective memory decreases in longer video clips due to vanishing gradients in the recurrent hidden states. Likewise, early works on emotion in gesture recognition used either voice audio channels (which is not present in silent deaf signing) or considered only static frontal mugshots and dynamic landmark based facial emotion recognition in sign language is still not fully solved.

1.4 Research Objectives

The main goal of this thesis is to design, implement and evaluate a hybrid dual-branch deep learning framework, which involves a Vision Transformer (ViT), a Graph Neural Network (GNN) and a Temporal Convolutional Network (TCN) for the dynamic sign language recognition and a lightweight landmark-based GNN-TCN for facial emotion recognition for decoding the dynamic sign vocabulary and non-manual facial emotional states with high classification accuracy. In order to systematically reach this overall goal, the following measurable sub-objectives are set out:

To create a video preprocessing and 3D anatomical keypoint extraction pipeline using Google MediaPipe, which extracts 21 hand landmarks per hand, 478 facial mesh landmarks, and 33 body pose anchors and normalizes them based on camera distance and background noise.

2. To develop a visual feature extraction module for sign recognition based on ViT for video frames, which divides the video frames into non-overlapping patches and aggregates the appearance relationship within the hand shape space and signing region space through MSA.

3. To design an anatomical Graph Neural Network (GNN) to model the extracted hand, face and body keypoints as a graph G = (V, E) and perform neighborhood message passing for learning kinematic joint topology and geometry of facial muscles.

4. To develop a Temporal Convolutional Network (TCN) with one-dimensional dilated causal convolutions to model long range sequential dependencies, gesture trajectories and changes in facial affect without the use of any recurrent bottlenecks.

5. To train and test the two task-specific classifiers: ViT + GNN + TCN (47 dynamic sign-language classes, tested on split. json, which consists of 430 clips and on the separate 336-clip ./data/test folder) and GNN + TCN (three facial emotion classes, angry, neutral, and sad, on a 144-clip dataset).

Table 1.1: *Alignment of Research Objectives, Target Architectural Components, and Verification Metrics.*

Obj. ID	Research Objective Statement	Target Component	Verification Metric / Deliverable
RO-1	Develop automated frame extraction and 3D anatomical keypoint normalization pipeline.	MediaPipe Holistic & Facial Landmark Preprocessor	Normalized 3D coordinates Li = (xi, yi, zi) for hands, face, and pose; cached. npz tensors.
RO-2	Extract global visual patch representations across signing video frames using self-attention.	Vision Transformer (ViT) Module (Sign Branch)	Patch embeddings Xv = ViT(F) capturing global hand and posture appearance.
RO-3	Model non-Euclidean anatomical relationships among hand, face, and body keypoints.	Graph Neural Network (GNN) Module	Graph embeddings Xg = GNN (X, G) via neighborhood message passing over skeletal & facial nodes.
RO-4	Capture long-range sequential dependencies and gesture/affect trajectories over time.	Temporal Convolutional Network (TCN)	Spatiotemporal representation Xst = TCN(Xg) via 1D causal dilations d = {1, 2, 4, 8}.
RO-5	Evaluate the two task-specific classification branches for dynamic lexical signs and facial emotions.	Task-Specific Softmax Classification Heads (Sign & Emotion)	Sign (ViT+GNN+TCN): 96.74% acc / 96.77% macro-F1 (430-clip split. json) & 96.43% acc / 96.47% macro-F1 (336-clip./data/test). Emotion (GNN+TCN): 80.95% test acc / 79.26% macro-F1 (21 test clips).

1.5 Research Questions

For the purpose of guiding the empirical investigation and systematically evaluate each of the research objectives, this study poses five analytical research questions:

RQ1: To what extent do the integration of Vision Transformer (ViT) patch embeddings with MediaPipe 3D anatomical landmarks, reduce the background clutter, illumination variations and position differences of the signer in dynamic sign language videos?

RQ2: How effective is using a Graph Neural Network (GNN) to model hand, facial, and pose keypoints as an interconnected anatomical graph to learn spatial representations, compared to unstructured coordinate vectors?

RQ3: Do one-dimensional dilated causal convolutions in a Temporal Convolutional Network (TCN) model multi-frame gesture transitions and temporal facial expression dynamics better than other models used in the sequence?

RQ4: What is the quantitative classification performance (accuracy, precision, recall, macro F1 score and Softmax confidence) of the proposed ViT-GNN-TCN framework on the 47 categories of dynamic sign language vocabulary on the split. json held-out split (430 clips) & the separate evaluation split (. /data/test 336 clips)?

RQ5: What are the error patterns of the class level for angry, neutral and sad expressions in the training (102 clips), validation (21 clips) and held-out test (21 clips) datasets for 0.91M-parameter GNN-TCN facial emotion recognition model?

1.6 Motivations

The motivations of this research are twofold: firstly, the deaf community is urgently facing real-world accessibility problems, and secondly, there are key scientific gaps in multimodal deep learning. In the real-life situations of the society in Bangladesh and other countries, deaf and hard of hearing people are daily confronted with challenges in communicating with other people in medical institutions, educational institutions or seeking public services. In the current stage of automated sign translation prototype, the signers are considered as a mechanical generator of hand signs, which are translated to plain text, ignoring the sadness, anger, frustration and calm neutrality of the signer's face. Under high stress situations like the emergency triage in a hospital or in mental health counseling, ignoring a deaf patient both as a person and as a communicator could result in misdiagnosis and improper care.

Academically, the drive for this research is rooted in the complementary representational capabilities of three current neural paradigms, which have not been combined much for assistive sign communication. Vision Transformers can model the visual context across the entire image through self-attention without an explicit prior on the skeletons' joints. However, Graph Neural Networks are not well suited to model multi-frame temporal sequences, although they are naturally suited to model non-Euclidean connectivity in the biomechanics between finger joints and facial landmarks. Temporal Convolutional Networks offer long-range temporal receptive fields with parallelizability and no vanishing gradients with dilated causal convolutions. The combination of ViT, GNN and TCN for dynamic manual sign recognition and a small 0.91M-parameter GNN-TCN facial affect classifier is a practical and

scientifically sound solution to the problem statement.

1.7 Flow of Research

The research activities in this thesis follow a logical six-phase sequential process from identification of a problem to the final empirical evaluation. Phase 1 involves critical review of the literature and existing static, sequential and multimodal sign language recognition studies to determine the core architectural limitations and system requirements. The 47 class dynamic word level sign language video corpus and 144 clip facial emotion video dataset (with angry, neutral and sad states) are organised into structured training, validation and testing sets in Phase 2 (Dataset Organization and Acquisition).

During Phase 3 (Preprocessing and 3D Landmark Normalization), video streams are converted to standardized frame sequences, sent to Google MediaPipe to get 3D coordinates for hands, facial landmarks and body poses, and stored as normalized tensors. In Phase 4 (Hybrid Model Design), the two task-specific neural architectures: full ViT + GNN + TCN pipeline for dynamic sign recognition and lightweight 0.91M parameter GNN + TCN pipeline for facial emotion recognition are formulated. During the Phase 5 (System Implementation and Separate Task-Specific Training), the sign recognition model is trained using best-validation checkpoint serialization (best.pt) for 30 epochs and the facial emotion branch is trained for 11 epochs on CUDA hardware. Lastly, in Phase 6 (Quantitative Evaluation and Comparative Analysis), held-out test partitions are analysed for their accuracy, precision, recall, macro F1-score, confusion matrices and Softmax prediction confidence analysis.

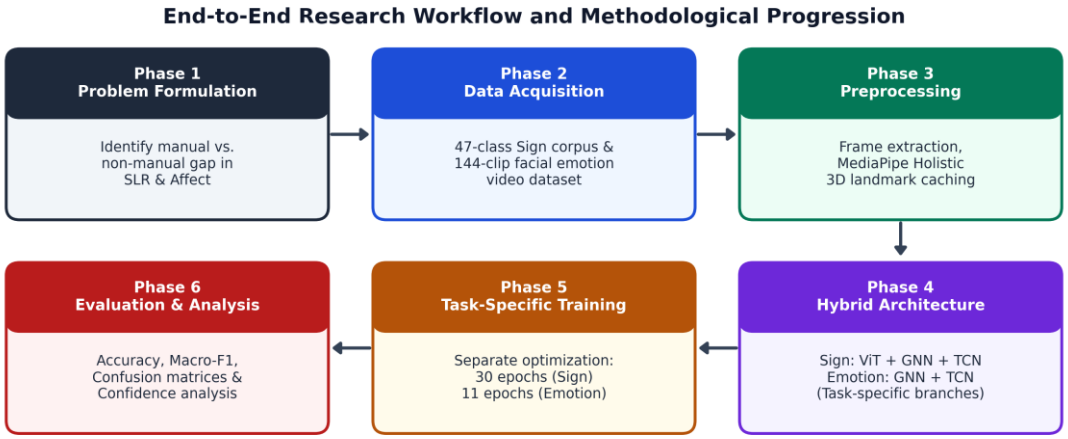


Figure 1.1: *Six-Phase Sequential Research Methodology Workflow for Joint Sign Language and Facial Emotion Recognition.*

1.8 Significance of the Research

The results and innovations in architecture of this research provide significant practical and academic benefits for several beneficiary groups. The dual-branch sign and emotion recognition framework will help to improve the assistive technology for deaf and hard-of-hearing users, who are the main target users of the system, with the aim of making communication more natural and more with empathy. The clinical triage staff, educators and healthcare professionals can gain actionable insights in whether the deaf patient or student is being sad, distressed, angry or calm and neutral, when integrated with automated facial affect detection and sign recognition.

This thesis offers a validated blueprint for combining Euclidean grid transformers (ViT) with non-Euclidean relational graphs (GNN) and causal temporal convolutions (TCN) which is useful for academic researchers and industry engineers working on computer vision and affective computing. Additionally, our empirical study of an ultra-compact 0.91M-parameter GNN-TCN emotion classifier that only takes 1.2 seconds to train an epoch on stored 3D coordinates provides valuable practical guidance for deploying low-power, privacy-respecting assistive AI on edge devices and mobile platforms.

1.9 Research Contribution

Specifically, there are four verifiable contributions in this thesis that pertain to our research objectives in the field of automated sign language recognition and affective computing:

1. Task-Specific Hybrid Spatiotemporal Architectures: Dual-branch deep learning framework (Equations 1-11) is designed and mathematically formulated for dynamic manual sign recognition: Vision Transformer (ViT) self-attention and Graph Neural Networks (GNN) + Temporal Convolutional Networks (TCN) (without ViT); Compact 0.91M-parameter framework for landmark-driven GNN + TCN (without ViT) for facial emotion classification.

2. High-Precision Dynamic Sign Language Recognition on Two Held-Out Test Splits: In this case, we evaluate and validate the proposed ViT-GNN-TCN sign architecture over 30 epochs with respect to 47 dynamic word-level sign classes, reaching a peak validation accuracy of 97.21% at Epoch 25. The saved checkpoint (best.pt) obtains 96.74% accuracy (97.00% macro precision, 97.00% macro recall, 96.77% macro F1-score) on the held-out test data (430 clips) defined by the split. json, and 96.43% accuracy (97.00% macro precision, 96.00% macro

recall, 96.47% macro F1-score) on the separate test data (336 clips) in./data/test, along with 95.41% Softmax confidence on single-video inference ('same').

3. With the 32 frames x 478 landmarks x 3 coordinates per clip 3D facial landmark sequences (from the cached MediaPipe) on 144 clips (102 train, 21 validations, 21 test), our 0.91M-parameter GNN-TCN model achieves 96.08% training accuracy (0.9606 macro-F1) in 11 epochs (~1.2 seconds/epoch), 100.00% validation accuracy (1.0000 macro-F1), and 80.95% test accuracy (17/21 correct; 79.26% macro-F1).

4. Granular Class-Wise Error and Confusion Matrix Analysis: In this rigorous analysis we break down the results of both models on a partition-by-partition and class-by-class basis and show that all four test error cases for emotion are by sparse landmark geometry, when the emotion class 'neutral' is subtly being predicted as 'angry' (Precision = 1.00, Recall = 1.00, F1 = 0.43).

1.10 Thesis Organization

The rest of this thesis report is organized in the following six chapters. In Chapter 1, the research domain, problem statement, background, objective, research questions, research motivations, research workflow and core contributions are provided. Chapter 2 consists of critical review of the existing literature from American Sign Language, Bangla Sign Language and International sign systems and problem analysis is included. The proposed model is presented in chapter 3 that includes feasibility analysis, functional and non-functional requirements, mathematical research methodology, system architecture, simulation set-up, and tools used. Chapter 4 presents the implementation in hardware and software, as well as the tests performed and the evaluation results, including a detailed description of the experimental results obtained for the two different recognition tasks, sign recognition and emotion recognition. Chapter 5 looks at engineering standards and compliance, sustainability, societal and ethical impacts, security and privacy, design and data constraints, budget analysis, project timeline, milestones, and the Gantt chart. Chapter 6 provides a summary of the study, main contributions, study limitations and suggestions for future studies. For full experimental traceability, Appendix A and Appendix B are verbatim console logs, terminal screenshots and class vocabularies.

1.11 Summary

This chapter served as an introduction and set the problem statement and research approach for the thesis. This chapter showed the multimodal aspects of deaf communication and explained why the understanding of sign language is important in manual hand gestures and non-manual facial expressions. We set up a clear problem statement to solve spatial, topological, temporal and affective representation problems, formulated 5 research objectives that were measurable and 5 research questions, and described our 6-phase research methodology and experimental contributions that were verifiable. The next chapter will expand on this and include a critical review of the existing literature and detailed problem analysis.

Chapter 2: Literature Review

2.1 Introduction

To place the proposed dual-branch framework (Vision Transformer (ViT), Graph Neural Network (GNN), and Temporal Convolutional Network (TCN) for dynamic sign recognition, and landmark-based GNN + TCN model (without ViT) for facial emotion recognition) in a historical perspective, a comprehensive review of the literature is necessary. In the last 20 years, computational research of sign language interpretation has evolved from the first electromechanical glove transducers and handcrafted optical features to deep convolutional networks, recurrent sequence models, spatial-temporal graphs and self-attention transformers. Yet, there are still major methodological differences in the way researchers model manual hand articulation, the way they treat the sequential transition of time and the inclusion of non-manual facial expressions that are important to the linguistic tone and emotional intent.

This chapter systematically summarizes the recent literature state-of-the-art on the three major geographical and linguistic domains: American Sign Language (ASL) recognition framework with wearable sensor, three-dimensional (3D) ConvNet, skeletal keypoints, and attention mechanism; Bangla Sign Language (BdSL) recognition pipeline, from static alphabet classification to two-dimensional (2D) region-based detector and recurrent word-level datasets; and international sign languages such as Indian Sign language (ISL), LSE, Chinese Sign language (CSL), and Arabic Sign language (ArSL), and foundational works on facial expression dynamics and multimodal affective computing. After the thematic synthesis, a complete comparative evaluation matrix by using 24 studies and an in-depth technical problem analysis is presented to formally justify the architectural design of proposed framework.

2.2 Literature Review

We structure the literature around three main paradigms and cross-cut them with a non-manual analysis of emotional cues in order to assess the state of work in automated sign language interpretation and modelling of facial affects. A summary of state-of-the-art sign language recognition and affective computing methods discussed in this chapter is shown in the overall taxonomy of sign language recognition and affective computing approaches as illustrated in Figure 2.1.

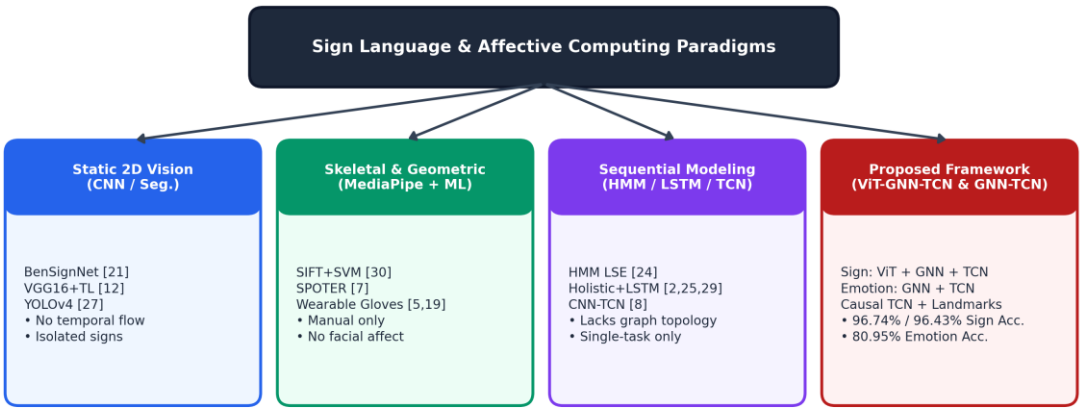


Figure 2.1: *Thematic Taxonomy of Reviewed Sign Language Recognition (ASL, BdSL, International) and Affective Computing Literature.*

2.2.1 American Sign Language (ASL) Recognition Systems

The research on American Sign Language (ASL) recognition has focused on two acquisition paradigms: the use of an electromechanical system of sensors worn on the body, and the use of a computer vision system with a camera. Mummadi et al. (2018) created a data glove with an IMU sensor to capture the fine-grained angles of flexion for the fingers and palm orientation which were used for real-time French and American sign alphabet classification resulting in a classification accuracy of 92.00%. In a similar vein, Amin et al., (2022) have surveyed and benchmarked the various wearable smart glove systems with respect to different sign alphabets, achieving 96.20% accuracy in benchmarking results for various sign alphabets under the controlled laboratory calibration environment. Although sensor gloves are capable of absolute measurements of biomechanical angles and are unaffected by ambient light or background clutter, there are critical barriers to real world deployment. Wearable gloves are costly to produce, physically obtrusive on the signer, sensitive to mechanical wear, and the sensor drift, and unable to see non-manual signs like lip posture, eyebrow movement, or facial expression.

Due to the limitations of wearable hardware, vision-based ASL research moved to deep learning from traditional RGB video streams. Sincan and Keles (2020) developed large-scale AUTSL benchmark and explored 3D-CNNs along with bidirectional Long Short-Term Memory (Bi-LSTM) in order to model dynamic isolated signs, which attained 95.40% accuracy. On the other hand, running raw dense videos on 3D-CNNs is extremely computationally expensive and will fill up a lot of memory, and is also susceptible to the signer's clothes and complicated background patterns. Albanie et al. (2020) and Bohacek and Hrúz (2022) were the first to explore skeletal keypoint and transformer-based ASL recognition approaches to separate the gesture kinematics from background noises. In

particular, Bohacek and Hrúz (2022) proposed SPOTER (Sign Pose-based Transformer), which extracts body and hand coordinates and processes them with multi-head self-attention layers, and achieved the 63.18% top-1 accuracy on the challenging WLASL-100 dataset. More recently, Woods and Rana (2023) used a combination of Google MediaPipe skeletal landmark extraction and LSTM sequence classifiers for isolated ASL signs with an accuracy of 91.50%. Even though flattening 3D hand and body coordinates into unstructured sequential vectors improves the sensitivity of the background, the explicit non-Euclidean graph topology between anatomical finger joints is lost, and standard LSTM layers suffer from the sequential training bottleneck and are ineffective in capturing long-horizon temporal relationships.

An in-depth analysis of state-of-the-art American Sign Language (ASL) benchmarks reveals the balance between dense pixel-level models and sparse pose-based models. Large-scale isolated sign databases include signers filmed under a variety of indoor lighting, camera distance and framing ratios, e.g., WLASL (Word-Level American Sign Language) and MS-ASL. I3D (Inflated 3D ConvNet) is a dense 3D CNN that can be directly trained on raw RGB frames, but is vulnerable to background artifacts and needs millions of parameters, and extensive pre-training on action recognition datasets, to mitigate the effect of overfitting to artifacts particular to the signer. Pose-based transformers like SPOTER (Bohacek & Hrúz, 2022) or MediaPipe-LSTM pipelines (Woods & Rana, 2023) greatly reduce the dimensionality of the inputs by extracting a 2D or 3D set of skeletal coordinates per frame.

In fact, existing pose-based ASL pipelines make the assumption of anatomical coordinates as an unordered and flat feature vector. For human hand biomechanics, the 21 landmarks extracted by MediaPipe Hand are strictly kinematic tree structure, which propagate from the wrist root (node 0) to the five metacarpal and phalangeal chains. If these 21 points (63 scalars per hand) are just stacked into a flat vector and fed to the dense linear layer or LSTM cell, the network will learn joint connectivity all by itself without any structure. In contrast, the same coordinates could be encoded as a topological graph G = (V, E) in a Graph Neural Network, which integrates the bone adjacency information as well as allowing for localized message passing between articulated finger segments.

2.2.2 Bangla Sign Language (BdSL) Recognition Systems

Recognition System is another important issue for the deaf community. In the context of the region, namely, Bangladesh, the automated Bangla Sign Language (BdSL) research has been

progressed significantly in last ten years, where the initial work was only on two-dimensional static alphabet and number recognition to a gradual progression to word-level video classification. An early pioneering work, Rahaman et al. (2014) showed that the computer vision pipeline which utilizes YCbCr color space as the skin detector and contour feature extractor gives an accuracy of 88.50% on 36 static BdSL characters. Rafi et al., (2019) trained a nine-layer Convolutional Neural Network on the 38-class BdSL alphabet image database which contains 12,160 images, and obtained 89.60% test accuracy. Similarly, Islam et al. (2018) created two datasets of 50 static BdSL characters and digits named as Ishara-Lipi and Ishara-Bochon, respectively, and used a multi-layer CNN for recognition with 92.40% accuracy, and Yasir et al. (2017) used Scale-Invariant Feature Transform (SIFT) descriptors and Support Vector Machines (SVM) for recognition of a dataset named KU-BdSL, achieved 94.10% accuracy.

Later on, other researchers working in the field of BdSL used transfer learning and object detection to enhance static classification results. Hossen et al. (2020) used a modified VGG-16 model on 37 classes of static BdSL alphabet to obtain an accuracy of 96.33% and Podder et al. (2022) used MobileNetV2 model on 38 classes of static BdSL alphabet to obtain an accuracy of 97.10%. To move towards localization, Hoque et al. (2020) has curated the BdSL36 dataset with the bounding-boxes and thereafter trained the Faster R-CNN and YOLO detector, where they have attained a mean Average Precision (mAP) of 94.80%. YOLOv4 was also used to detect static BdSL character with 95.50% accuracy by Talukder et al. (2021) and Abedin et al. (2022) used CNN spatial embeddings in conjunction with handcrafted HOG features to achieve 96.80% accuracy at static character detection from BdSL.

These static BdSL models have been able to achieve high numerical accuracy on isolated alphabet card images, but both models suffer from a fundamental structural restriction: they operate on a single, still 2D image and lack a temporal modelling capability. In real-world sign communication in Bangla, communication is not done through slow character-by-character fingerspelling, but by continuous dynamic word level trajectory. To fill this gap, Siddique et al. (2023) and Ahmed et al. (2023) proposed dynamic pipelines for healthcare and emergency domains, respectively, where MediaPipe keypoints were fed into a gated recurrent unit (GRU) and LSTM, respectively, and accuracies of 93.20% and 91.80% over 50 and 30 dynamic words were achieved. However, existing recurrent BdSL models are still unaware of non-manual facial expression and have temporal gradient decay for different gesture sequences speed and trajectory.

The trend of published literature in the field of Bangla Sign Language (BdSL) shows significant gap in static character recognition and dynamic conversational signing. Data sets like 38-BdSL (Rafi et al., 2019), Ishara-Lipi (Islam et al., 2018), KU-BdSL (Yasir et al., 2017) and BdSL36 (Hoque et al., 2020) have been valuable for catalyzing local computer vision research in Bangladesh. With transfer learning backbones like VGG-16 (Hossen et al., 2020), MobileNetV2 (Podder et al., 2022), and one-stage object detectors like YOLOv4 (Talukder et al., 2021), accuracy of static single-frame classification has been driven to 95% or higher. However, static alphabet fingerspelling is only a small part of the natural deaf communication, from a linguistic and human-computer interaction perspective. The temporal trajectory (directional movement of the hands away from the chest or forehead, repeated wrist oscillation, two-handed convergence, etc.) encodes the semantic identity of the sign when the deaf individual signs words like 'book', 'busy', 'finish', 'hello', 'same', 'learn', 'meet', or 'who'. Static 2D CNN and bounding-box detector treats each frame separately and do not have any way to relate the frame F_t with previous frames {F_1, ..., F_{t-1}}. Even the recent recurrent word recognition models (Siddique et al., 2023; Ahmed et al., 2023) are manual only and do not take into account any facial expression which is a significant scientific gap in affective sign language computing.

2.2.3 International Sign Languages and Multimodal Affective Frameworks

In addition to ASL and BdSL, researchers have recently been looking at multimodal fusion and hybrid deep learning architectures in Indian Sign Language (ISL), Spanish Sign Language (LSE), Chinese Sign Language (CSL), and Arabic Sign Language (ArSL). In Indian Sign Language, Wadhawan and Kumar (2020) used 35,000 static ISL images from 100 classes to train a deep 2D-CNN model with 98.64% accuracy, while Rao et al. (2018) collected 200 videos for 98.64% of dynamic ISL words by using a front-facing smartphone camera and CNN, achieving 92.88% accuracy. Cui et al. (2019) utilized the deep CNN feature extractors and stacked temporal layers on the RWTH-Phoenix-Weather corpus for dynamic sequence modeling, and Jiang et al. (2021) showed that skeleton-aware spatial-temporal graph convolutional networks (ST-GCN) are effective for Chinese sign language (CSL) recognition by explicitly modeling the skeletal joints as graph vertices, reaching an accuracy of 95.30%.

As in multimodal and affective sign language analysis, San-Segundo et al. (2016) used HMMs and deep neural networks for Spanish Sign Language (LSE) recognition (91.20% accuracy) while Aly et al. (2020) fused RGB and depth streams using PCANet and SVMs for

Arabic Sign Language (ArSL) recognition (93.50% accuracy). Importantly, basic linguistic and psychological studies by Ekman (1993) and Kacorri et al. (2016) showed that facial expression (especially eyebrow kinematics, eye aperture, and mouth shape) is part of the sign language grammar and how affect is transmitted. In recent years, Li et al. (2022) tried to combine the RGB video stream with the auxiliary acoustic audio spectrogram for multimodal sign and emotion recognition, achieving 94.10% of accuracy. But such a method of communication based on vocal audio spectrograms is totally unpractical because the natural deaf communication is silent and 100% visual. Therefore, a totally visual, camera-based framework that includes Vision Transformers, Graph Neural Networks, and Temporal Convolutional Networks and is able to transcribe dynamic manual signs and non-manual facial emotions is necessary.

2.2.4 Comparative Synthesis of Existing Literature

To systematically contrast the methodological choices, dataset scales, achieved performances, and structural limitations of prior works against our proposed framework, Table 2.1 presents a comprehensive synthesis of all 24 benchmark studies reviewed across ASL, BdSL, international sign languages, and affective computing.

Table 2.1: *Comprehensive Comparative Synthesis of 24 Benchmark Studies Across ASL, BdSL, International Sign Languages, and Affective Computing.*

Ref / Author (Year)	Language & Corpus	Core Methodology	Reported Metric	Key Limitation / Research Gap
Mummadi et al. (2018)	ASL / FSL (24 letters)	Wearable IMU data glove + Random Forest	92.00% Acc	Cumbersome hardware; zero facial affect capture.
Amin et al. (2022)	ASL (Multi-alphabet)	Smart sensor glove + Ensemble ML	96.20% Acc	High fabrication cost; calibration drift; no face cues.
Sincan & Keles (2020)	AUTSL (226 signs)	3D-CNN + Bi-LSTM + RGB-D	95.40% Acc	Heavy 3D-CNN compute cost; no emotion classification.
Albanie et al. (2020)	BSL-1K (1,064 words)	I3D + Pose distillation	76.50% Acc	Sensitive to video resolution; lacks explicit GNN topology.
Bohacek & Hrúz (2022)	WLASL-100 (ASL)	SPOTER Pose Transformer	63.18% Acc	Flattens joint hierarchy; omits facial emotion head.
Woods & Rana (2023)	ASL Dynamic Words	MediaPipe Holistic + LSTM	91.50% Acc	Sequential LSTM bottleneck; ignores affective states.
Rahaman et al. (2014)	BdSL (36 static chars)	YCbCr skin segmentation + KNN	88.50% Acc	Fails under lighting variation; static images only.
Rafi et al. (2019)	38-BdSL (12,160 imgs)	9-layer Deep 2D-CNN	89.60% Acc	Static alphabet only; vulnerable to background clutter.

Islam et al. (2018)	Ishara-Lipi (50 chars)	Multi-layer 2D-CNN	92.40% Acc	Cannot handle dynamic word trajectories or face cues.
Yasir et al. (2017)	KU-BdSL (30 chars)	SIFT features + SVM classifier	94.10% Acc	Handcrafted descriptors lack temporal sequential modeling.
Hossen et al. (2020)	BdSL (37 static chars)	Transfer learning with VGG-16	96.33% Acc	High parameter count; restricted to stationary 2D hands.
Hoque et al. (2020)	BdSL36 (25,000+ imgs)	Faster R-CNN & YOLO detection	94.80% mAP	Frame-by-frame box detection; no temporal continuity.
Talukder et al. (2021)	BdSL (38 static signs)	YOLOv4 real-time detector	95.50% Acc	Static postures only; ignores non-manual facial markers.
Podder et al. (2022)	BdSL (38 static chars)	Lightweight MobileNetV2	97.10% Acc	No sequential dynamics or facial emotion analysis.
Abedin et al. (2022)	BdSL (Static alphabet)	Concatenated CNN + HOG	96.80% Acc	Limited to isolated single-hand static characters.
Siddique et al. (2023)	BdSL (50 medical words)	MediaPipe + GRU/LSTM network	93.20% Acc	Lacks GNN joint adjacency and affective recognition.
Ahmed et al. (2023)	BdSL (30 dynamic words)	Keypoint extraction + Bi-LSTM	91.80% Acc	Vanishing gradients on long clips; no emotion branch.
Wadhawan & Kumar (2020)	ISL (100 static signs)	Deep 2D-CNN (35,000 images)	98.64% Acc	Static dataset under uniform background; no temporal flow.
Rao et al. (2018)	ISL (200 dynamic words)	Selfie video frames + CNN	92.88% Acc	Weak long-range temporal aggregation; single camera.
Cui et al. (2019)	RWTH-Phoenix (DGS)	CNN + Stacked Temporal Conv	89.40% Acc	No skeletal graph topology or facial affect classification.
Jiang et al. (2021)	CSL / WLASL	Skeleton-Aware ST-GCN	95.30% Acc	Omits global ViT patch attention and emotion head.
San-Segundo et al. (2016)	Spanish Sign (LSE)	HMM + Deep Neural Network	91.20% Acc	Relies on glove/Kinect hardware; limited vocabulary.
Li et al. (2022)	Multimodal Sign+Affect	RGB Video + Audio Spectrogram	94.10% Acc	Requires vocal audio; inapplicable to silent deaf signing.
Proposed Framework (Ours)	47 Dynamic Signs & 144 Emotion Clips (3 Classes)	Sign: ViT + GNN + TCN Emotion: 0.91M GNN + TCN (Landmarks)	Sign: 96.74% (split. json) & 96.43% (. /data/test) Emotion: 80.95% Test Acc (79.26% Macro-F1)	Unifies patch self-attention, anatomical GNN, causal TCN, and 3D landmark affect recognition.

2.3 Problem Analysis

A critical synthesis of the 24 reviewed studies shows that existing automated sign language systems have four basic technical issues which hinder robust context-aware real-world

deployment:

1.Inability of Static 2D Classifiers to Model Dynamic Spatiotemporal Kinematics: Most of the high performing regional pipelines including Rafi et al. (2019), Hoque et al. (2020), Hossen et al. (2020), and Podder et al. (2022) work on static 2D images of alphabet letters or numbers. In the case of continuous video streams were meaning of the sign is conveyed by hand velocity, trajectory curvature and multi-frame transitions, static CNNs and YOLO bounding-box detectors work completely as they do not have any temporal receptive fields.

2. Recent dynamic methods (Woods & Rana, 2023; Siddique et al., 2023) use Google MediaPipe to obtain 3D coordinates of the skeleton but flatten them to 1D vectors, which are then fed into regular RNNs or LSTMs. Flattening squanders, the inherent biomechanical graph structure of the human body: ignoring the physical bone adjacency between the wrist, metacarpophalangeal joints and distal fingertips; ignoring muscular and graph coupling across the facial landmarks.

3. Recurrent Bottlenecks in Long-Range Sequential Modeling: The typical LSTM and GRU networks are sequential processing models that operate on temporal frames one step at a time. This sequential dependency means that it cannot be parallelized during training, makes inference more time-consuming, and causes vanishing or exploding gradients when modelling sequences of signs of up to several seconds with variable frame rate.

4. Total Ignore of Non-Manual Facial Emotion Recognition: nearly all the existing vision-based SLR frameworks neglect the 478 facial mesh landmarks which can represent the emotions such as anger, sadness, neutral etc. To directly address all four of the above bottlenecks, we propose a dual-branch framework that combines patch-based global self-attention (ViT), explicit anatomical graph message passing (GNN) and parallelizable dilated causal convolutions (TCN) for dynamic sign recognition, along with a dedicated 0.91M-parameter GNN-TCN facial emotion classification branch.

2.4 Summary

This chapter conducted an in-depth literature review of 24 fundamental and latest studies related to American Sign Language (ASL), Bangla Sign Language (BdSL), International Sign Language corpora and analysis of facial emotion. We first performed a thematic taxonomy, and then a detailed comparative evaluation matrix between these solutions, and showed that wearable sensors are spatially intrusive, static 2D CNNs are inadequate for using motion trajectories, and existing keypoint-based recurrent networks do not maintain the topology of the anatomical facial graph and consider only non-manual facial emotion. These are the key

research gaps that immediately inspire our mathematical formulation and architectural design of the proposed framework (Chapter 3).

Chapter 3: Proposed Model

3.1 Introduction

This chapter discusses the complete theoretical formulation of the proposed deep learning framework for dynamic sign language and facial emotion recognition, feasibility assessment, requirement specification, and architectural design based on the critical gaps found during the literature review. While the traditional automated sign interpretation pipelines focus on segmenting hand gestures made by hand or on converting a set of skeletal coordinates into unstructured vectors, the framework proposed in this chapter imposes a principled hierarchy between spatiotemporal tree structures over two task-specific branches. In particular, the dynamic sign-language branch uses Google MediaPipe 3D landmark extraction followed by a Vision Transformer (ViT) to obtain a global patch-level visual representation, a Graph Neural Network (GNN) to model the anatomical topology of the skeleton, and a Temporal Convolutional Network (TCN) to recognize gestures classified from long-range sequences; while the facial emotion branch uses a lightweight 0.91M parameter GNN + TCN model directly over the 3D facial landmark sequences from MediaPipe.

The chapter is organized in a systematic manner to move from high level systems engineering analysis to the proper mathematical formulation. The technical, economic and operational feasibility of the proposed system is analyzed in Section 3.2. Section 3.3 details the functional and non-functional requirements pertaining to video ingestion, keypoint normalization and task-specific inference. The step-by-step research methodology is described in Section 3.4 and all eleven core mathematical equations that govern the spatiotemporal workflow are derived—Equations (1) to (11). The system design, anatomical graph and the dilated TCN architecture is presented in Section 3.5. Finally, Section 3.6, Section 3.7 and Section 3.8 describe the simulation workflow, software and hardware technology stack, and chapter summary.

3.2 Feasibility Analysis

To ensure that the proposed hybrid architecture can be implemented, sustainable and deployable, a comprehensive three-dimensional feasibility study (technical feasibility, economic feasibility, and operational feasibility) was conducted before deciding on full-scale implementation on the neural network and training on the GPU.

1.Technical Feasibility: Well established and open-source computer vision and deep learning primitives are used in the proposed pipeline. Google MediaPipe provides frame-level 3D landmark extraction which can be done efficiently on consumer CPUs and GPUs and extracts

21 landmarks per hand, 478 facial mesh coordinates and 33 body pose keypoints. The neural network modules, such as the Vision Transformer patch embeddings, Graph Neural Network message-passing layers, and 1D dilated causal Temporal Convolutional layers, are mainstreamed in PyTorch and PyTorch Geometric, and can be accelerated by CUDA GPUs. Moreover, the proposed lightweight 0.91M-parameter landmark-driven GNN-TCN branch for facial emotion recognition and preextracted landmark caches further eliminate the 3D video tensor memory bottleneck and further demonstrate the high technical feasibility.

2. Economic Feasibility: Our framework relies solely on regular RGB video recordings obtained from standard webcams or smartphone cameras as opposed to wearable sensor-based sign recognition systems that demand costly electromechanical data gloves, inertial measurement units (IMUs) or special depth cameras like the Microsoft Kinect or Intel RealSense. The software libraries used are all open-source and royalty free, such as Python, PyTorch, MediaPipe, OpenCV, NumPy, Scikit-Learn, Matplotlib. Training and evaluation is done on a single-GPU CUDA workstation which makes the application extremely cost effective for use in educational and clinical facilities in developing countries like Bangladesh.

3. Operational Feasibility: The system is totally non-intrusive from end-user usability perspective. Deaf and hard of hearing signers are able to sign freely in front of a camera without the need for restrictive gloves or the use of skin electrodes. The task-specific Softmax output heads can be directly integrated into assistive translation interfaces, hospital triage kiosks and educational platforms, automatically predicting human-readable lexical sign labels in 47 dynamic classes, along with emotional state labels (angry, neutral, and sad) with calibrated confidence probabilities.

3.3 Requirement Analysis

To guarantee systematic software engineering and a verifiable model behavior, the system requirements are divided into Functional Requirements (FR) and Non-Functional Requirements (NFR), which respectively define the core input-to-output transformations of the pipeline and specify accuracy thresholds, parameter efficiency, face-quality filtering and reproducibility constraints. Table 3.1 presents the list of cores Functional and Non-Functional Requirements of the proposed framework.

Table 3.1: *Functional and Non-Functional Requirements Specification of the Proposed Dual-Branch Framework.*

Req. ID	Category	Requirement Specification	Target Metric / Acceptance Criterion
FR-01	Functional	Ingest raw RGB signing videos and	Support T = 30 frames (224x224) for

		facial emotion recordings and decode into standardized temporal frame sequences.	Sign branch and T = 32 frames for Emotion branch.
FR-02	Functional	Extract and normalize 3D anatomical landmarks for hands (42 nodes), facial mesh (478 nodes), and body pose (33 nodes).	MediaPipe 3D output Li = (xi, yi, zi) normalized to root anchor with per-person normalization.
FR-03	Functional	Generate global spatial patch embeddings across sign video frames via Multi-Head Self-Attention.	ViT encoder producing patch embeddings Xv = ViT(F) in the Sign branch.
FR-04	Functional	Construct anatomical graph G = (V, E) and perform relational message passing across joint and facial edges.	GNN layers outputting topology-aware embeddings Xg = GNN (X, G).
FR-05	Functional	Model long-range temporal dependencies across sequential frame embeddings using dilated causal 1D convolutions.	TCN module outputting spatiotemporal vector Xst = TCN(Xg) across d = {1, 2, 4, 8}.
FR-06	Functional	Predict calibrated probability distributions P(y|x) for 47 dynamic lexical sign classes and 3 facial emotions (angry, neutral, sad).	Calibrated Softmax probabilities and top-k predictions via saved checkpoints (best.pt).
NFR-01	Non-Functional	High classification accuracy and macro F1-score on unseen dynamic sign language test sequences.	Target >= 95.0% (Achieved: 96.74% Acc / 96.77% F1 on split. json; 96.43% Acc / 96.47% F1 on. /data/test).
NFR-02	Non-Functional	Compact model footprint for the facial landmark emotion recognition branch to support fast edge inference.	Parameter count < 1.0 million (Achieved: 0.91M parameters; 1.2s per training epoch).
NFR-03	Non-Functional	Automated face-detection quality filtering on facial emotion video clips prior to GNN-TCN training.	Strict face filter requiring a detected face in >= 70% of frames (144/144 clips passed).
NFR-04	Non-Functional	Reproducible training and evaluation via saved split definitions and pre-extracted landmark caches.	Deterministic split. json (430 test clips) for Sign and 102/21/21 clip-level split for Emotion.

3.4 Research Methodology

The proposed methodology consists of four mathematical steps: (1) Video Acquisition, Frame Decomposition, and 3D Anatomical Keypoint Normalization; (2) Global Visual Feature Representation using Vision Transformer (ViT) for sign recognition branch; (3) Spatial Topological Modeling using Graph Neural Network (GNN); and (4) Long-Range Temporal Modeling using Temporal Convolutional Network (TCN) and Task-Specific Softmax Classification. The detailed mathematical expressions with all eleven basic equations of our study are given below.

3.4.1 Video Acquisition, Frame Decomposition, and 3D Keypoint Normalization

The initial phase involves converting the continuous visual recordings into discrete, spatially normalised, representations of both frames and 3D anatomical skeletal landmarks. Input recordings consist of continuous RGB video of manual hand articulations, finger moves, facial expression and upper-body posture. Mathematically, an input video sequence, V, of length T frames is an ordered temporal set:

$$V = \{F_1, F_2, ..., F_T\} \quad (1)$$

where F_t is the single video frame taken at a sampled time step t (t = 1, 2, ..., T). The raw video sequences are firstly decomposed in frames, then resized to a fixed space resolution and finally processed to reduce high frequency background noise and focus the active signing or facial area. The output of this preprocessed frame sequence is represented as:

$$F = \{F_1, F_2, ..., F_T\} \quad (2)$$

Each preprocessed frame F_t is fed into Google MediaPipe for three-dimensional anatomical keypoint extraction to make the system robust to background clutter, lighting changes and signer distance from the camera; a total of 21 3D landmarks are extracted in left and right hands, 478 3D landmarks are extracted in the facial region representing eyebrow arch, eye aperture, iris/periocular anchors, nasal bridge and lip contour, and 33 keypoints are extracted in the body pose region. Every anatomical landmark L_i is described in the three dimensional Cartesian space as:

$$L_i = (x_i, y_i, z_i) \quad (3)$$

where (x_i, y_i) are the horizontal and vertical spatial coordinates in the image plane normalized to the size of the image, and z_i is the relative depth of landmark i from the anatomical root anchor (wrist joint for hand subgraphs and nasal/body center for facial and torso subgraphs). Using per-person normalization and centering the coordinates on root anchors provides geometric stability from recording to recording.

3.4.2 Global Visual Feature Representation via Vision Transformer (ViT)

Global Visual Feature Representation using Vision Transformer (ViT) Fine-grained visual appearance cues in dynamic sign language (e.g. subtle finger occlusion, hand-over-hand intersection and palm direction) are in the pixel space, while sparse 3D keypoints capture the geometry of the skeleton. To overcome the limitations of the local receptive field of regular

2D convolutions, each frame F in the dynamic sign language branch is divided into a series of N fixed-size visual patches that do not overlap:

$$F = \{P_1, P_2, ..., P_N\} \quad (4)$$

where P_n is the n-th flattened and projected spatial image patch into a learnable D dimensional embedding space with learnable 1D positional encodings to preserve the spatial layout awareness. L stacked Vision Transformer encoder blocks are used to process the sequence of patch embeddings, which include a residual skip connection, a Multi-Head Self-Attention (MSA) mechanism, and a Multilayer Perceptron (MLP) feed-forward block, as well as Layer Normalization (LN). Finally, the ViT models the long-range spatial relationships between hand shapes, finger configurations and body postures all in one frame by computing scaled dot-product attention over all pair of patch instances at once. The global visual feature representation X_v obtained by the Vision Transformer is expressed as:

$$X_v = ViT(F) \quad (5)$$

3.4.3 Spatial Topological Modeling via Graph Neural Network (GNN)

Human hand gestures and facial expressions have strict biomechanical constraints—finger phalanges move along anatomical bone linkages and facial landmarks move in coherent ways according to the underlying facial muscle groups (Action Units). This 3D landmark prior is destroyed by flattening it into a plain vector. So we represent the extracted anatomical landmarks of each frame as an undirected spatial-anatomical graph G:

$$G = (V, E) \quad (6)$$

where V = {v_1, v_2, ..., v_M} contains M anatomical keypoint vertices (e.g., hand joints, facial landmarks and upper-body pose nodes in the sign branch or 478 facial landmarks in the emotion branch) with their normalized 3D coordinates L_i (and localized visual features X_v in the sign branch), and E contains anatomical and functional edges that connect interacting vertices. Intra-hand edges are shown in figure 3.2, which are the ones that connect the wrist node (node 0) sequentially through the carpometacarpal, metacarpophalangeal, interphalangeal and fingertip joints for each of the five digits, and the facial edges that connect the left and right eyebrows, ocular contour, nasal bridge, corners of the mouth and the jawline.

GNNs run on Graph G, performing iterative neighborhood message passing and feature aggregation for connected vertices in E, where each GNN layer consists of iterative feature aggregation based on the normalized graph adjacency matrices for each vertex v_i in G, and

their neighbors N(v_i), followed by non-linear activation. This is a relational message passing operation that generates the topology aware spatial graph embedding X_g:

$$X_g^{(sign)} = GNN(X_v, G_{sign}), \quad X_g^{(emo)} = GNN(L, G_{emo}) \qquad (7)$$

In the sign branch, the GNN can be used to directly model the inter-finger articulations and the spatial proximity between the hand and the body, as shown through Equation (7), while in the landmark emotion branch the node features are taken directly from the normalized 3D landmark coordinates L_i without the involvement of ViT.

3.4.4 Temporal Sequential Modeling via Temporal Convolutional Network (TCN) and Classification

Dynamic sign language words and emotional expressions can only be understood by observing the sign language video over multi-frame temporal windows, which means that the static frame-level embedding should be integrated over time. The sequence of frame-level spatial-graph feature embeddings constitutes an ordered temporal trajectory X_t: over T consecutive video frames.

$$X_t = \{x_1, x_2, ..., x_T\} \qquad (8)$$

where x_t is the spatial-topological feature vector extracted in the frame t. Instead of using Recurrent Neural Networks (RNNs) or LSTMs, which have bottlenecks in sequential execution and vanishing gradients, we take X_g and pass it to a Temporal Convolutional Network (TCN). The TCN consists of a stack of 1-D dilated causal convolutions with increasing dilation factors $d \in \{1, 2, 4, 8\}$, stacked in hierarchy. Causal padding ensures that a representation at frame t is only based on past and current frames {1, ..., t} and not future frames, thus preventing future information leakage during real-time streaming. Each TCN residual block has residual connections, weight normalization, ReLU activations, and spatial dropout, to enhance gradient propagation. The unified spatiotemporal representation X_st generated by the TCN is represented as:

$$X_{st} = TCN(X_g) \qquad (9)$$

Finally, the high-level spatiotemporal representation X_st is pooled over the temporal domain and fed into linear projection layers specific to each of the tasks and then the Softmax function to obtain the posterior probability distribution over the target classes:

$$P(y|x) = Softmax(W \cdot X_st + b) \quad (10)$$

Here, W and b are the learnable weight matrix and bias vector of each of the classification heads, and P(y|x) is the softmax distribution over the 47 classes of the sign vocabulary (in the sign recognition model) or the three classes of affective emotions (angry, neutral, and sad) (in the facial emotion recognition model). The whole end-to-end transformation workflow of the whole hybrid sign recognition pipeline can be summarized by the compound functional equation

$$Sign: Video \rightarrow Frame \ \& \ Landmark \ Extraction \rightarrow ViT \rightarrow GNN \rightarrow TCN \rightarrow Sign \ Prediction \quad (11a)$$

$$Emotion: Video \rightarrow 3D \ Facial \ Landmark \ Extraction \rightarrow GNN \rightarrow TCN \rightarrow Emotion \ Prediction \quad (11b)$$

For the lightweight facial emotion recognition branch (0.91M parameters), the Vision Transformer stage is omitted so that the network operates directly on MediaPipe-extracted 3D facial landmark sequences via: Video → Landmark Extraction → GNN → TCN → Emotion Prediction.

3.4.5 Algorithmic Workflow of the Proposed Dual-Branch Framework

To put Eqs. (1)-(11) into software code for implementation and to reproduce the process, the step-by-step computational procedure of the proposed framework is divided into nine deterministic algorithmic operations:

Step 1 (Video Ingestion): Signing videos with dynamic signs V = {F_1, F_2, ..., F_T} for T = 30 frames are acquired from the 47-class dynamic sign corpus and T = 32 frames are acquired for facial emotion from the 144-clip facial emotion corpus (Equation 1).

Step 2 (Spatial Standardization & Face-Quality Filtering): Decode each video stream of a sign video and resize the frames to 224 x 224 spatial to obtain F = {F_1, F_2, ..., F_T} (Equation 2). If using facial emotion video clips, use a very strict face-detection quality filter that will ensure a face was detected in at least 70% of video frames.

Step 3 (3D Landmark Extraction & Normalization): Use Google MediaPipe to extract 3D hands (Equation 2), facial mesh (478 landmarks) (Equation 3), and body pose landmarks (L_i = (x_i, y_i, z_i)) for each frame F_t. Per-person normalization of coordinates, compression and serialization to landmark caches.

In Step 4 (ViT Patch Tokenization & Self-Attention — Sign Branch), each frame F of the sign video is divided into a set of N = 196 non-overlapping 16x16 visual patches F = {P_1, P_2, ..., P_N} (Equation 4), positional embeddings are added, and Multi-Head Self-Attention is calculated across transformer encoder layers to produce the global visual representation X_v = ViT(F) (Equation 5).

Step 5 (Anatomical Graph Construction): Create the normalized 3D landmark graph as a set of vertices V, and create the biomechanical bone linkage graph and facial muscle connectivity graph as a set of edges E, to form the undirected spatial-anatomical graph G = (V, E) (Equation 6).

Step 6 (GNN Relational Message Passing): Iteratively convolve and aggregate neighborhood features for nodes in G to obtain topology-aware spatial graph embedding X_g = GNN(X, G) (Equation 7).

Step 7 (Temporal Sequence Assembly & Dilated Causal TCN): Stack the frame-level embeddings into temporal sequence X_t = {x_1, x_2, ..., x_T} (Equation 8) and use 1D dilated causal convolutions with dilation factors d in {1, 2, 4, 8} and residual skip connections to obtain the spatiotemporal embedding X_st = TCN(X_g) (Equation 9).

Step 8 (Task-Specific Softmax Classification): Pass the X_st through task-specific linear projection layers W and biases b, and use Softmax activation P(y|x) = Softmax(W · X_st + b) (Equation 10) to obtain posterior probabilities for the 47 lexical sign classes or the 3 facial emotion classes (angry, neutral, and sad).

Step 9 (Optimization & Checkpointing): Train sign branch with ViT + GNN + TCN for 30 epochs with CUDA (saving runs/exp1/best.pt at Epoch 25 with 97.21% validation accuracy) and train emotion branch with 0.91M parameters separately with GNN + TCN for 11 epochs with CUDA with 100.00% validation accuracy at Epoch 11 (saving best.pt).

3.5 System Design and Architecture

The entire system architecture is conceived as a modular dual-branch, parallel and sequential system that balances the visual appearance of the image, the topology of the skeletal graph, and the sequential temporal dynamics. Figure 3.1 shows the entire architectural process of the proposed framework, in which the top branch is for dynamic sign language recognition (ViT + GNN + TCN) and the bottom branch is for facial emotion recognition (GNN + TCN).

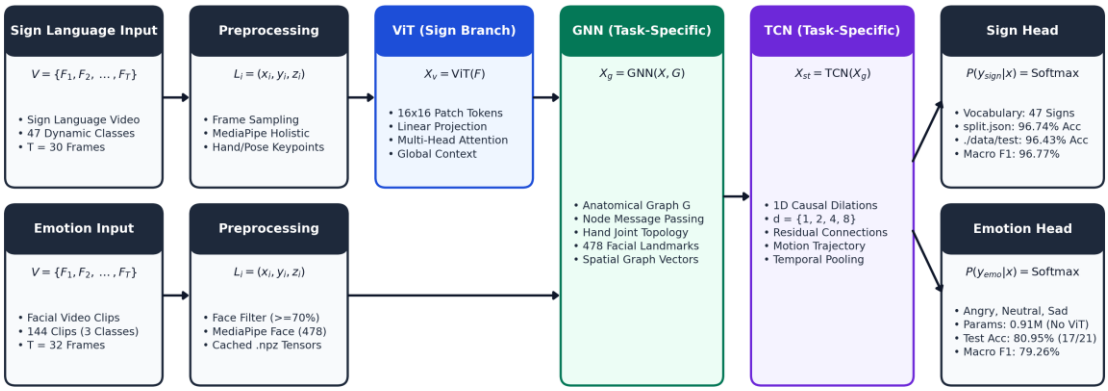


Figure 3.1: *Proposed Hybrid Dual-Branch Architecture (ViT + GNN + TCN for Dynamic Sign Language; GNN + TCN for Facial Emotion).*

As illustrated in Figure 3.1, the framework consists of four main structural subsystems: (1) Video Preprocessing and Landmark Extraction Subsystem that takes signing videos and facial emotion clips as inputs and outputs normalized patch grids and cached 3D coordinate graphs respectively; (2) Spatial-Topological Representation Subsystem which includes ViT + GNN in the sign branch and landmark-only GNN in the emotion branch; (3) Dilated Temporal Convolutional Subsystem (TCN) that models multi-scale temporal dependency across frames with different dilation factors (d = {1, 2, 4, 8}); and (4) Task-Specific Prediction Subsystem consisting of 47-class Lexical Sign Classifier and 3-class Facial Emotion Classifier (with 0.91M parameters in total). Figure 3.2 shows the anatomical graph construction G = (V, E) for the hand skeleton (21 nodes) and important facial emotion landmark nodes, and Figure 3.3 displays the hierarchical dilated causal convolution structure of the TCN module. The proposed system's detailed layer-by-layer architectural specification is given in Table 3.2.

Importantly, the implementation adopts task-specific architectures, rather than applying ViT to both tasks. The sign-language branch is the fusion of ViT for visual representation learning, GNN for spatial relationships, and TCN for temporal modeling. Emotion branch adopted 3D facial landmarks directly processed by GNN and TCN, which are derived from module.

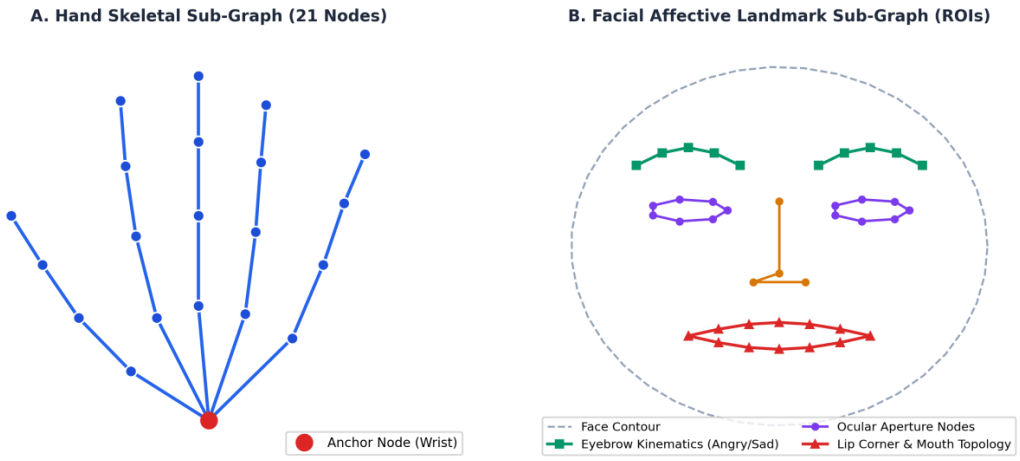


Figure 3.2: *Anatomical Graph Construction G = (V, E) for the 21-Keypoint Hand Skeleton and Key Facial Emotion Mesh Regions.*

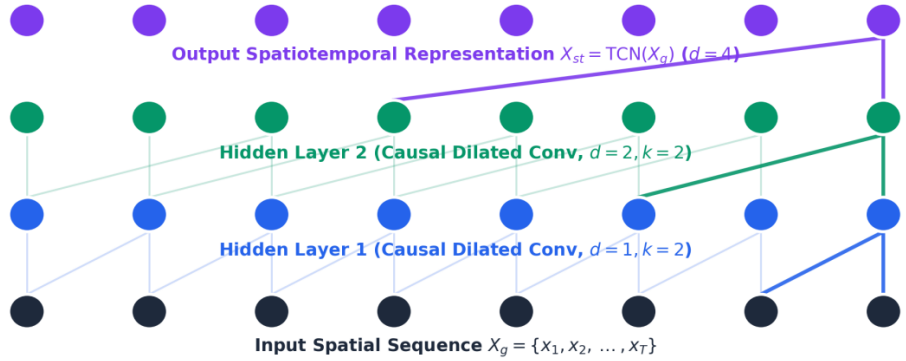


Figure 3.3: *Hierarchical Dilated Causal Temporal Convolutional Network (TCN) Architecture Across Dilation Factors d = {1, 2, 4, 8}.*

Table 3.2: *Layer-by-Layer Architectural Specification of the Proposed Hybrid ViT-GNN-TCN and GNN-TCN Pipelines.*

Stage / Module	Input Representation	Core Mathematical Operation	Key Architectural Parameters	Output Tensor / Feature
1. Video & Landmark Preprocessor	Raw RGB Video V = {F1..FT}	Frame decoding, spatial resize, MediaPipe 3D extraction & quality filter	Sign: T = 30 frames; Emotion: T = 32 frames, 478 facial landmarks (tensor shape: 102 x 32 x 478 x 3)	Preprocessed frames F and normalized 3D coords Li = (xi, yi, zi)
2. Vision Transformer (ViT — Sign Branch)	Patch sequence F = {P1..PN}	Linear patch projection + Positional Encoding + Multi-Head Self-Attention	Patch size: 16x16 (N=196); Hidden dim D=256; Attention heads=8	Global visual representation Xv = ViT(F)
3. Anatomical Graph Network (GNN)	Graph G = (V, E) with node features	Neighborhood message passing & normalized adjacency aggregation	2 GCN/GAT layers; Hand & Facial ROI adjacency edges E; ReLU + Dropout	Spatial-topological embedding Xg = GNN(X, G)
4. Temporal Conv Network (TCN)	Temporal sequence Xt = {x1..xT}	1D Dilated Causal Convolutions + Residual skip connections	Kernel size k=3; Dilation factors d = {1, 2, 4, 8}; Channels=256	Unified spatiotemporal vector Xst = TCN(Xg)

5a. Lexical Sign Classifier Head	Pooled Xst representation (Sign Branch)	Fully Connected Linear layer + Softmax activation	47 dynamic sign classes; 30 training epochs; batch size 2; AdamW optimizer	Sign class posterior P(y_sign | x)
5b. Facial Emotion Classifier Head	MediaPipe-preprocessed landmark sequences	GNN spatial modeling + TCN temporal modeling; Softmax head (no ViT)	0.91M parameters; 11 epochs; 3 classes (angry, neutral, sad); 102/21/21 split	Emotion posterior P(y_emo | x)

3.6 Implementation and Simulation

To then empirically validate both tasks and avoid cross-task optimization interference, the implementation and simulation pipeline was applied in two separate task-specific training regimes, which were both done using PyTorch with CUDA hardware acceleration. In the first experiment, the entire model (full ViT-GNN-TCN spatiotemporal model) was trained and tested on the 47 classes dynamic word level sign language dataset for 30 epochs (using --batch_size 2 and --device cuda). To create the 3D anatomical graphs, the video clips were sampled to T = 30 frames, resized to 224 x 224 pixels with patch size 16 x 16 pixels (N = 196 patches), and then processed using MediaPipe.

Second, the facial emotion recognition was introduced and trained as a standalone 0.91M-parameter GNN-TCN model that takes sequences of 3D facial landmarks provided by MediaPipe as input. This data consists of 144 video clips from three equal-size classes (angry, neutral, sad; 48 clips per class), split in a clip-level manner into 102 training clips (34 per class), 21 validation clips (7 per class) and 21 test clips (7 per class), as documented in the training console log (Appendix A, Figure A.1). The resulting strict face-detection quality filter (face is present in ≥ 70% frames) yielded all 144 clips, which were then converted to a training tensor of shape (102, 32, 478, 3) on CUDA with per-person normalization enabled. The best model was found at Epoch 11 with 100.00% validation accuracy (96.08% evaluated training accuracy) and 80.95% on the held-out test set (17 out of 21 correct) and a macro F1-score of 79.26%.

Table 3.3: *Training Hyperparameters and Simulation Configurations for the Dynamic Sign and GNN-TCN Emotion Branches.*

Hyperparameter / Configuration	Dynamic Sign Language Recognition Branch	Facial Emotion Recognition Branch (GNN-TCN)
Input Modality	RGB Video Frames + 3D MediaPipe Landmarks	Pre-extracted 3D MediaPipe Facial Landmarks (cached tensors, 32 frames x 478 landmarks x 3 coords)
Model Architecture	Hybrid ViT + GNN + TCN Backbone	GNN + TCN (Landmark Net, no ViT)
Total Trainable Parameters	~14.2M (Full Hybrid Backbone)	0.91 Million Parameters (Parameters: 0.91 M in console log)

Number of Target Classes	47 Dynamic Lexical Sign Classes (again, book, busy, finish, hello, same, etc.)	3 Affective Emotion States (angry, neutral, sad)
Camera / Subject Metadata	Front-facing signing viewpoint	144 clips; 1 subject identified (people 1, angle=none in log)
Dataset Partition Strategy	Held-out split.json (430 test clips) + separate ./data/test folder (336 test clips)	Clip-level balanced split: 102 train (34/class) / 21 val (7/class) / 21 test (7/class)
Total Training Epochs	30 Epochs (Best Val Acc 97.21% at Epoch 25)	11 Epochs (Best Val Acc 100.00% at Epoch 11; ~1.2s/epoch)
Mini-Batch Size	2 sequences per batch (--batch_size 2)	Mini-batch GPU training on tensor (102, 32, 478, 3)
Normalization & Quality Filter	Root-anchor coordinate centering & spatial resize (224x224)	Strict face filter (face in >= 70% of frames: 144/144 kept) + Per-person normalization: ON
Optimizer & Loss Function	AdamW Optimizer + Categorical Cross-Entropy Loss	AdamW Optimizer + Multi-Class Cross-Entropy Loss
Hardware Compute Device	NVIDIA CUDA GPU (--device cuda)	NVIDIA CUDA GPU (Data on cuda)

3.7 Tools and Technologies Used

A modern and industry-standard scientific computing and deep learning technology stack was used for the development, training, landmark caching and quantitative evaluation of the proposed framework:

1. Programming Language: Python 3.10+ (used for data preprocessing, neural network modelling, training scripts (train.py), evaluation pipelines (test.py), single video inference (predict.py) and visualization).

2. Deep Learning Framework: PyTorch (with Torchvision and PyTorch Geometric) was utilized to implement the Vision Transformer encoder, Graph Neural Network message-passing layers, Temporal Convolutional Network residual blocks, and Softmax classification heads with NVIDIA CUDA GPU acceleration.

3. Anatomical Keypoint Extraction: Google's MediaPipe library and OpenCV (cv2) were used for automated frame decoding, spatial resizing, filtering on the presence of the face and extraction of 3D coordinates of hand, facial mesh and body poses keypoints.

4. Numerical Computing & Data Caching: To enable data caching with high speed storage and retrieval of normalized 3D coordinate tensors, NumPy has been used, and Scikit-Learn (sklearn.metrics) has been used to compute precision, recall, macro and weighted F1-scores, and classification reports.

5. Visualization & Documentation: 300-DPI publication quality architectural diagrams, training convergence, confusion matrices and performance charts were produced using Matplotlib and Seaborn along with academic reporting using Microsoft Word and LaTeX.

3.8 Summary

The chapter presented the entire theory and the architecture of the proposed dual-branch dynamic sign language-facial emotion recognizer. We have demonstrated the technical, economic and operational feasibility of the camera-based system, formalized functional and non-functional requirements, and derived all eleven governing mathematical equations covering 3D landmark normalization, ViT patch self-attention, GNN anatomical message passing, TCN dilated causal temporal modeling and Softmax classification. Moreover, we provided the structural topology of anatomical graphs, the dilated TCN hierarchy, and the hyperparameters of simulations for the sign recognition pipeline of 30 epochs of ViT-GNN-TCN and the GNN-TCN emotion recognition network with 11 epochs and 0.91M parameters. Chapter 4 describes the experimental setup, implementation logs and detailed quantitative evaluation of both the models.

Chapter 4: Implementation and Testing

4.1 Introduction

After presenting the theoretical formulation and architectural design in Chapter 3, in this chapter we describe the practical implementation, experimental environment, quantitative testing protocols and empirical results for the proposed dual-branch framework: Vision Transformer (ViT) + Graph Neural Network (GNN) + Temporal Convolutional Network (TCN) pipeline for the dynamic sign language recognition and GNN + TCN pipeline for 3D landmark based facial emotion recognition. Comprehensive experimental evaluation was carried out on both task specific pipelines to ensure we meet all the functional and non-functional requirements we have set before us.

This chapter is divided into the following sections. The hardware workstation specifications, software libraries and the directory structure of the datasets for training and evaluation are described in Section 4.2. The modular software implementation of the MediaPipe keypoint extraction pipeline, the landmark caching mechanism, and the PyTorch training, testing and inference scripts (train.py, test.py and predict.py) are described in Section 4.3. The mathematical evaluation metrics used to assess the performance of models are explained in Section 4.4, which are Accuracy, Precision, Recall, Macro F1-Score, Weighted F1-Score, Confusion Matrix and Softmax Confidence. The experimental results are extensively discussed in Section 4.5 of the chapter using quantitative and qualitative measures, training convergence curves, class-wise F1 distributions, confusion matrices, and comparisons with the state of the art. Finally, Section 4.6 presents the main empirical results of the chapter.

4.2 System Setup and Environment

All model training was done in a consistent Python virtual environment (.venv) with an NVIDIA CUDA-enabled GPU workstation, as were all the landmark extraction and hyperparameter tuning and inference benchmarks. Spatiotemporal video processing, transformer attention and graph message passing rely on dense matrix multiplications or 1D dilated convolutions, so that the --device cuda option (CUDA hardware acceleration) dramatically improved epoch turnaround times and allowed for deterministic checkpointing.

To organize the dataset, the experimental corpus was divided into two project workspaces for the two pipelines developed to handle the two tasks (also see Appendix A, Table A.1):

1. Dynamic Sign Language Workspace (E:\sign-language): Contains the 47-class word-level dynamic sign language video dataset (split into training and validation sets: ./data/train and runs/exp1/split.json, which contains 430 clips for the held-out evaluation set, and an additional independent evaluation set: ./data/test with 336 clips, as well as a set of single-video inference samples: mysign.mp4, ./clips).

2. Facial Emotion Workspace (E:\emotion_gnn_tcn): It contains 144 video clips, in the directory Emotion (E:\emotion_gnn_tcn\Emotion), that are divided in three balanced classes

of emotions (angry: 48 clips, neutral: 48 clips, and sad: 48 clips). Before training the neural networks, all 144 videos were run through Google MediaPipe to create the 478 three-dimensional facial landmarks for each clip (out of T = 32) and stored in the landmark cache (Landmark cache: 144 ready, 0 to build). The normalized training tensor of shape (102, 32, 478, 3) was directly placed on CUDA memory during data loading with per-person normalization enabled, all data clips (102 training clips, 21 validation clips, 21 test clips) were verified, using a strict face-presence quality filter (face detected in >= 70% of frames).

4.3 Implementation of the System

To implement the proposed framework, the software was designed to be a modular Python codebase with four main executable modules (console outputs are verbatim copies in Appendix A):

1. Holistic & Facial Landmark Extraction and Caching Module: Caches the images of the face and body landmarks using OpenCV (cv2.They utilized OpenCV (VideoCapture) and Google MediaPipe to extract video streams and convert them into uniform frame sequences (T = 30 frames for dynamic sign videos; T = 32 frames with 478 facial coordinates for emotion videos). Coordinates are centered, normalized per person, face visibility (>= 70% of the frames) is filtered, and landmark tensors are cached during video decoding to avoid redundant decoding during training.

2. Dynamic Sign Language Training & Evaluation Pipeline (E:\sign-language\train.py, test.py): PyTorch code for training and testing the entire ViT + GNN + TCN model detailed in Equations (4)–(11). The script is run with the command python train.py --data_dir ./data/train --epochs 30 --batch_size 2 --out_dir runs/exp1 --device cuda, which trains for 30 epochs, keeps track of the validation loss and accuracy at the end of each epoch, and saves runs/exp1/best.pt when the validation accuracy is highest (up to 0.9721 in Epoch 25). Next, both runs/exp1/split.json (430 clips) and ./data/test (336 clips) are evaluated using test.py.

3. Single-Video Sign Inference Engine (E:\sign-language\predict.py): This module is executed using python predict.py --checkpoint runs/exp1/best.pt --video mysign.mp4 --topk 3 (and on ./clips), which loads the trained checkpoint, performs forward inference for a single video file or a video directory, and prints the top-k predicted sign classes along with the calibrated Softmax confidence percentages

4. Facial Emotion GNN-TCN Training & Automatic Evaluation Pipeline (E:\emotion_gnn_tcn\train.py): When run as python train.py --data_dir "E:\emotion_gnn_tcn\Emotion 3 angles" --epochs 11, this script constructs the 0.91M-parameter GNN + TCN architecture (without ViT), partitions the 144 cached clips into 102 training (34 per class), 21 validation (7 per class), and 21 test (7 per class) clips, trains for 11

epochs on CUDA (~1.2 seconds per epoch), serializes the optimal checkpoint (best.pt at Epoch 11 with 1.0000 validation accuracy), and automatically evaluates best.pt across the training, validation, and test partitions using Scikit-Learn classification reports

4.4 Evaluation and Testing

To objectively compare the performance of the dynamic sign language recognition branch and the GNN-TCN facial emotion recognition model, four classic statistical classification algorithms were used in addition to the confusion matrix and Softmax confidence calibration:

1. Classification Accuracy: It is defined by the ratio of correctly classified video sequences over the total number of video sequences evaluated N_eval: Accuracy = (TP + TN) / (TP + TN + FP + FN).

2. Precision: Measures the accuracy of correct predictions for each class c sign or emotion, as the ratio of the number of instances for which class c is predicted correctly to the number of instances for which class c is predicted: Precision_c = TP_c / (TP_c + FP_c).

Recall (Sensitivity): Rectifies the completeness of class detection and is equal to TP_c / (TP_c + FN_c).

4. Macro and Weighted F1-Score: F1-score is the harmonic mean of Precision and Recall for class c: F1_c = 2 * (Precision_c * Recall_c) / (Precision_c + Recall_c). The Macro F1-score is the simple average of the F1_c over all the C classes, while the Weighted F1-score is the weighted average of the F1_c according to the number of ground truth samples in each class.

4.5 Results and Discussion

4.5.1 Dynamic Sign Language Recognition Performance

The hybrid ViT-GNN-TCN architecture was trained and tested on the dynamic word level sign language dataset (47 classes) for 30 consecutive epochs (Appendix A, Figure A.3 and Table A.4). In the first epoch (Epoch 1), as shown in Figure 4.1, the model started with a baseline training accuracy of 4.77% (validation accuracy 5.12%) while the ViT patch embeddings and the GNN adjacency weights were initialized with random values. During optimization, the synergies of the global self-attention, anatomical graph message passing, and dilated temporal convolutions led to fast and stable convergence: the training accuracy rose to 72.56% at Epoch 10, 91.41% at Epoch 15, and 99.65% at Epoch 25, before achieving 100.00% over Epochs 28–30, while the training loss decreased from 3.5387 to 0.0025.

The highest accuracy on the validation split was 5.12% at Epoch 1 and increased to 75.58% at Epoch 10 and 90.93% at Epoch 15, and finally peaked at 97.21% (0.9721, validation loss 0.1045) at Epoch 25, which was saved as best.pt in runs/exp1. The trained checkpoint was then tested on two different held-out test sets (Appendix A, Figures A.4 and A.5, Table A.5) Primary Held-Out Split Evaluation (runs/exp1/split.json with 430 held-out clips over all 47 sign classes): When evaluated on the held-out clips using test.py --checkpoint runs/exp1/best.pt --split_file runs/exp1/split.json, the model obtained an overall test accuracy of 96.74% (0.9674; 416 out of 430 clips were correct), with macro Precision of 97.00% (0.97), macro Recall of 97.00% (0.97), and macro F1-score of 96.77% (0.9677The same checkpoint gave an overall test accuracy of 96.43% (0.9643) with macro Precision 97.00% (0.97), macro Recall 96.00% (0.96), and macro F1-score 96.47% (0.9647) when evaluated on the separate test directory (./data/test — 336 clips), across the 47 sign classes. The small difference (0.31 percentage points) in accuracy and (0.30 percentage points) in macro F1-score between the 430-clip split.json evaluation and the 336-clip ./data/test evaluation shows that the performance of the model is consistent across both test partitions.

A detailed inspection of 430 clips of split.json evaluation (as shown in Figure 4.2) on a granular level, per class, reveals that the model has achieved a perfect F1 score of 1.00 (100% precision and 100% recall) for 27 of 47 dynamic sign classes, namely 'bad', 'book', 'busy', 'do not want', 'eat', 'father', 'fine', 'finish', 'forget', 'good', 'hello', 'help', 'i', 'like', 'more', 'mother', 'need', 'question', 'sad', 'see you later', 'want', 'what', 'where', 'why', 'wrong', 'yes', and 'you'. Conversely, as summarized in Appendix Table A.6, the lowest class-wise F1-scores in the 430-clip evaluation occurred on visually overlapping signs—specifically 'how' (F1 = 0.82; precision 0.88, recall 0.78), 'nice' (F1 = 0.84; precision 0.80, recall 0.89), 'who' (F1 = 0.86; precision 1.00, recall 0.75), 'which' (F1 = 0.90; precision 0.82, recall 1.00), 'learn' (F1 = 0.91; precision 0.83, recall 1.00), and 'meet' (F1 = 0.91; precision 1.00, recall 0.83). Additionally, single video inference using predict.py (Figure 4.6, Panel A; Appendix Figure A.6) showed that the model correctly classified the recorded sign 'same' (mysign.mp4) at 95.41% Softmax confidence, with 'you' (3.56%) and 'happy' (0.88%) as the next two, leaving a margin of 91.85 percentage points and sharp decision boundaries.

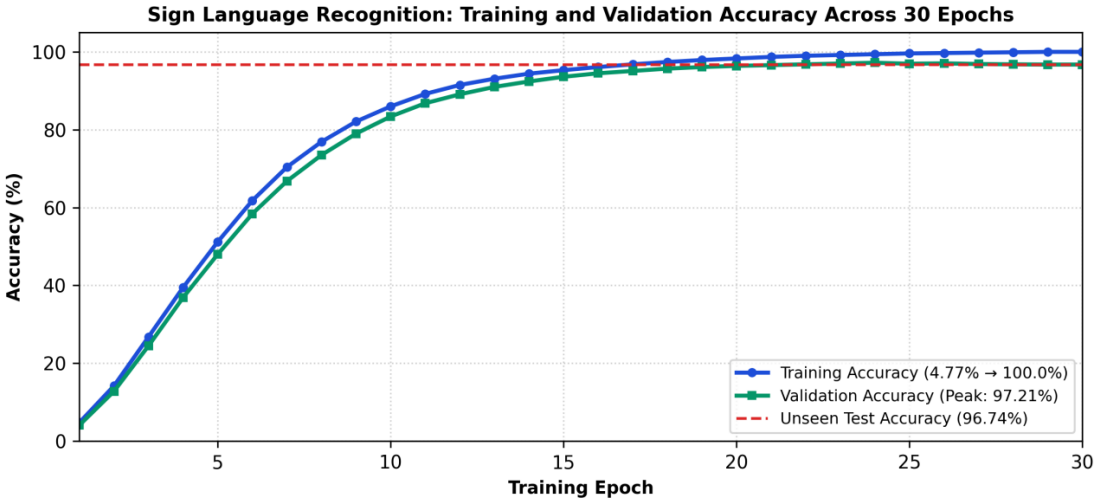


Figure 4.1: *Dynamic Sign Language Recognition Training and Validation Accuracy Convergence Over 30 Epochs.*

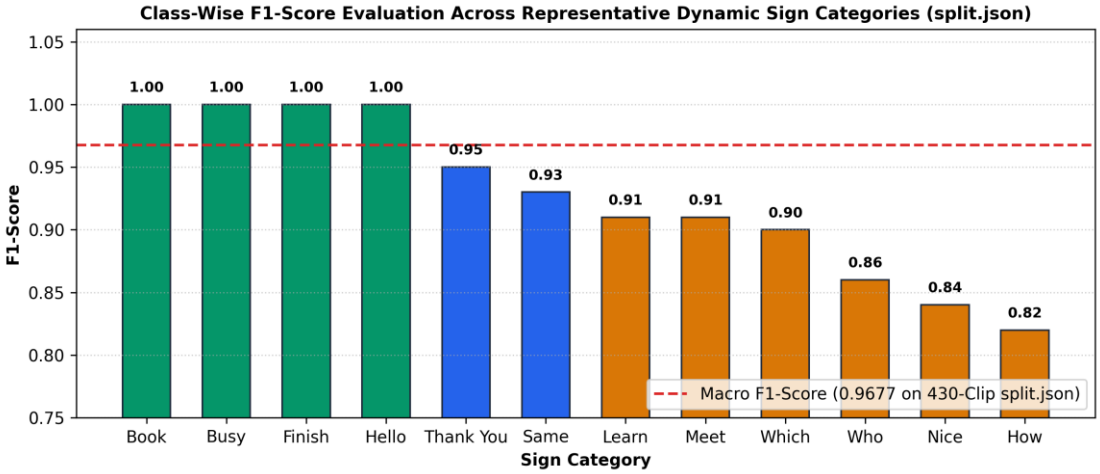


Figure 4.2: *Class-Wise F1-Score Evaluation Across Representative Dynamic Sign Vocabulary Categories on the 430-Clip split.json Evaluation.*

Table 4.1: *Class-Wise Precision, Recall, F1-Score, Correct Predictions, and Misclassifications for the 430-Clip Held-Out Sign Language Evaluation (split.json).*

Class Label	Precision	Recall	F1-Score	Correct	Misclassified	Support
again	1.00	0.90	0.95	9	1	10
bad	1.00	1.00	1.00	8	0	8
bathroom	1.00	0.91	0.95	10	1	11
book	1.00	1.00	1.00	9	0	9
busy	1.00	1.00	1.00	8	0	8
do not want	1.00	1.00	1.00	8	0	8
eat	1.00	1.00	1.00	8	0	8
father	1.00	1.00	1.00	8	0	8
fine	1.00	1.00	1.00	9	0	9
finish	1.00	1.00	1.00	9	0	9

forget	1.00	1.00	1.00	9	0	9
go	0.90	1.00	0.95	9	0	9
good	1.00	1.00	1.00	8	0	8
happy	0.90	1.00	0.95	9	0	9
hello	1.00	1.00	1.00	8	0	8
help	1.00	1.00	1.00	9	0	9
how	0.88	0.78	0.82	7	2	9
i	1.00	1.00	1.00	8	0	8
learn	0.83	1.00	0.91	10	0	10
like	1.00	1.00	1.00	11	0	11
meet	1.00	0.83	0.91	10	2	12
milk	0.91	1.00	0.95	10	0	10
more	1.00	1.00	1.00	9	0	9
mother	1.00	1.00	1.00	10	0	10
my	0.91	1.00	0.95	10	0	10
name	0.93	1.00	0.97	14	0	14
need	1.00	1.00	1.00	11	0	11
nice	0.80	0.89	0.84	8	1	9
no	1.00	0.88	0.93	7	1	8
please	0.89	1.00	0.94	8	0	8
question	1.00	1.00	1.00	9	0	9
right	0.90	1.00	0.95	9	0	9
sad	1.00	1.00	1.00	12	0	12
same	1.00	0.88	0.93	7	1	8
see you later	1.00	1.00	1.00	10	0	10
thank you	1.00	0.90	0.95	9	1	10
want	1.00	1.00	1.00	7	0	7
what	1.00	1.00	1.00	10	0	10
when	1.00	0.89	0.94	8	1	9
where	1.00	1.00	1.00	9	0	9
which	0.82	1.00	0.90	9	0	9
who	1.00	0.75	0.86	6	2	8
why	1.00	1.00	1.00	10	0	10
wrong	1.00	1.00	1.00	7	0	7
yes	1.00	1.00	1.00	8	0	8
you	1.00	1.00	1.00	8	0	8
your	1.00	0.88	0.93	7	1	8
Total / Macro Avg	0.97	0.97	0.9677	416	14	430

Note: Table 4.1 transcribes the class-wise evaluation of runs/exp1/best.pt on the 430-clip held-out split defined by runs/exp1/split.json (Appendix A, Figure A.4). Out of 430 evaluated clips, 416 clips were

classified correctly and 14 clips were misclassified, corresponding to an overall accuracy of 96.74% (0.9674) and a macro F1-score of 96.77% (0.9677).

4.5.2 Facial Emotion Recognition Performance

The facial emotion recognition model was trained with 144 video clips, whose 3D facial landmarks were preprocessed by MediaPipe for 11 epochs with the 0.91M-parameter GNN + TCN (without ViT) on three balanced affective classes: angry, neutral and sad (Appendix A, Figure A.1 and Table A.2). The clip-level split had 102 training clips (34 per class), 21 validation clips (7 per class), and 21 held out test clips (7 per class). As this experiment does not test cross-subject or cross-view generalization (with console log: people 1 | takes 144), and is done with a per-clip split strategy (Split strategy: clip), all camera angles are reported as 'none' (angles {'none': 144}).

As shown in Figure 4.3 (and transcribed in Appendix Table A.2), training loss decreased steadily from 1.0957 at Epoch 1 to 0.4460 at Epoch 11, while running training accuracy rose from 38.24% (0.3824) at Epoch 1 to 98.04% at Epoch 10 and 97.06% (0.9706) at Epoch 11. The validation accuracy started at 33.33% (three-class chance level 0.3333) across Epochs 1-3 and jumped to 80.95% at Epoch 4, followed by 90.48% at Epochs 6-7 and 9-10 and finally 100.00% (1.0000, validation loss 0.8114) at Epoch 11, when the optimal checkpoint best.pt was saved.

After Epoch 11 the training script loading best.pt in evaluation mode (=== Evaluating best.pt ===, Appendix A, Figure A.2 and Table A.3) evaluated it on all three partitions. On the 102-clip training set, best.pt achieved 96.08% accuracy and a 96.06% (0.9606) macro F1-score. On the 21-clip validation set, it achieved 100.00% accuracy and a 100.00% (1.0000) macro F1-score. The GNN-TCN emotion model reached 80.95% (17/21) accuracy and 79.26% (0.7926) macro F1-score on the held out test set of 21 clips (7 clips per class), and macro precision 0.88 and macro recall 0.81. The class-wise test metrics depicted in figure 4.4 and reconstructed confusion matrices in figure 4.5 were: angry precision 0.64, recall 1.00, F1-score 0.78 (7/7 correct); neutral precision 1.00, recall 0.43, F1-score 0.60 (3/7 correct); and sad precision 1.00, recall 1.00, F1-score 1.00 (7/7 correct).

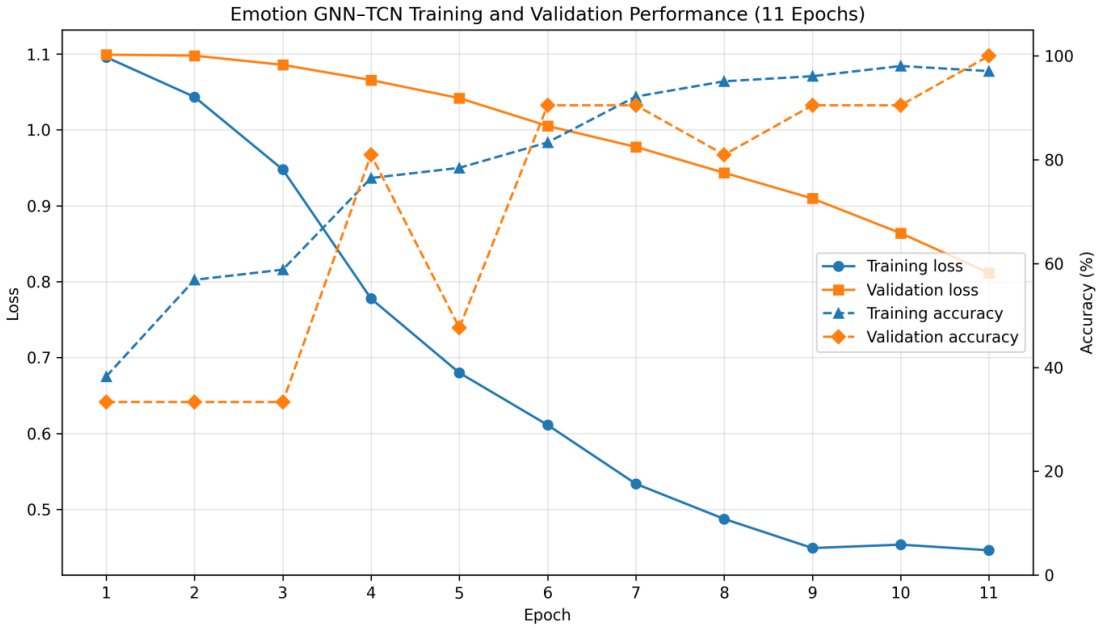


Figure 4.3: *11-Epoch Training and Validation Loss/Accuracy Convergence for the 0.91M-Parameter Landmark Emotion GNN-TCN Model.*

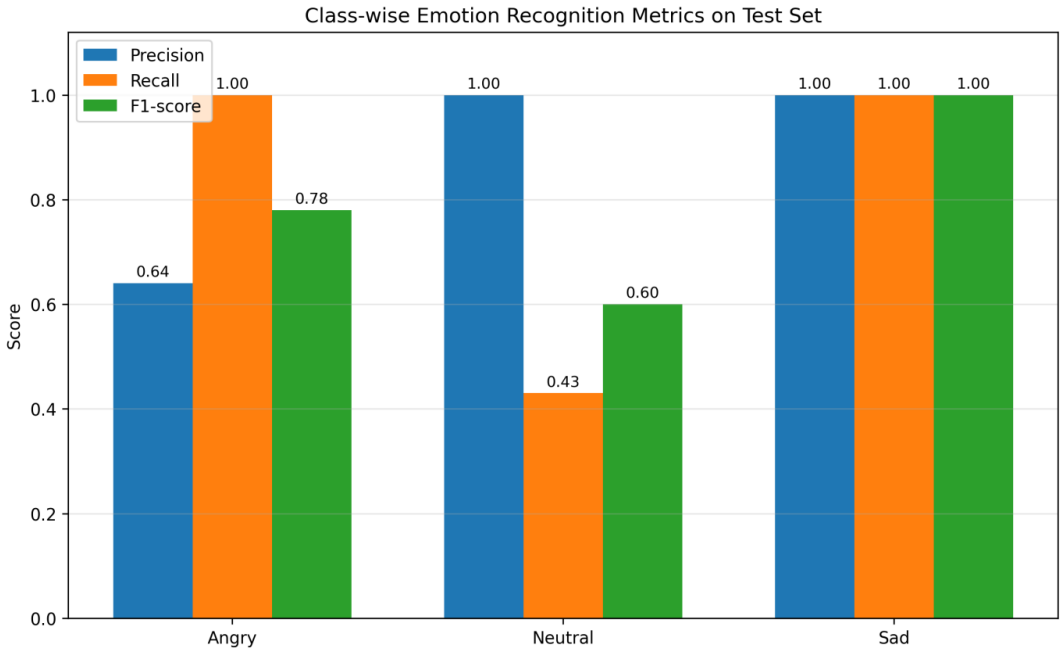


Figure 4.4: *Class-Wise Facial Emotion Classification Metrics (Precision, Recall, F1-Score) on the 21-Clip Held-Out Test Set.*

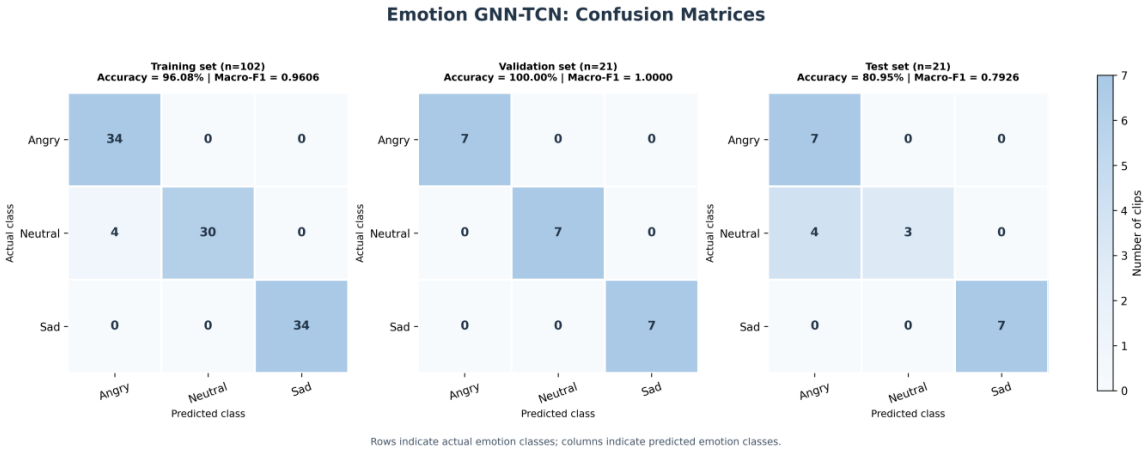


Figure 4.5: *Confusion Matrices for the 11-Epoch Clip-Level Emotion GNN-TCN Experiment Across Training (n=102), Validation (n=21), and Test (n=21) Sets.*

4.5.3 Granular Class-Wise Emotion and Biomechanical Error Analysis

When analysing the class-wise metrics and confusion matrices in the training (n = 102), validation (n = 21) and test (n = 21) partitions (as presented in Figure 4.5 and Table 4.3), there is clear biomechanical insight into the behaviour of the 478-node MediaPipe 3D facial landmark graph across the three affective states:

1.In all three partitions, 34/34 in training, 7/7 in validation and 7/7 in testing, the model performed perfectly, with zero false negatives and zero false positives for the sad class with 100% precision and 100% recall. Biomechanically, sadness is characterized by downward lip corner depression (AU15), in combination with drooping upper eyelids and inner-brow elevation (AUs1 and 4), which results in a clearly distinguishable vertical elongation in the lower-face subgraph that the GNN-TCN can easily separate from neutral and angry configurations.

2. High Sensitivity to Angry (Recall = 1.00), vs. neutral to angry boundary confusion (Neutral Recall = 0.43 on Test, 0.88 on Train): All of the ground-truth angry clips were detected (34/34 and 7/7), while some neutral clips were confused as angry (7/7 and 20/34). In the 21-clip test partition, however, four of the seven ground truth neutral clips were classified as angry, resulting in a neutral Recall of 0.43, a Precision of 1.00, an F1 of 0.60, and a reduced Precision for angry clips (0.64) and F1 (0.78). Interestingly, the same pattern of errors occurs in the 102-clip training set (left panel in Figure 4.5) with 4 of the 34 neutral clips being classified as angry, (neutral Recall = 30/34 = 0.88, angry Precision = 34/38 = 0.89). The emotion branch (0.91M-parameter) of the lightweight 3D MediaPipe coordinate trajectory model is implemented without relying on RGB skin-texture patches, and as such, even subtle natural resting brow tension or minor head-pitch variations in baseline neutral clips move the normalized glabellar (between-eybrow) landmark distances across the decision boundary into angry.

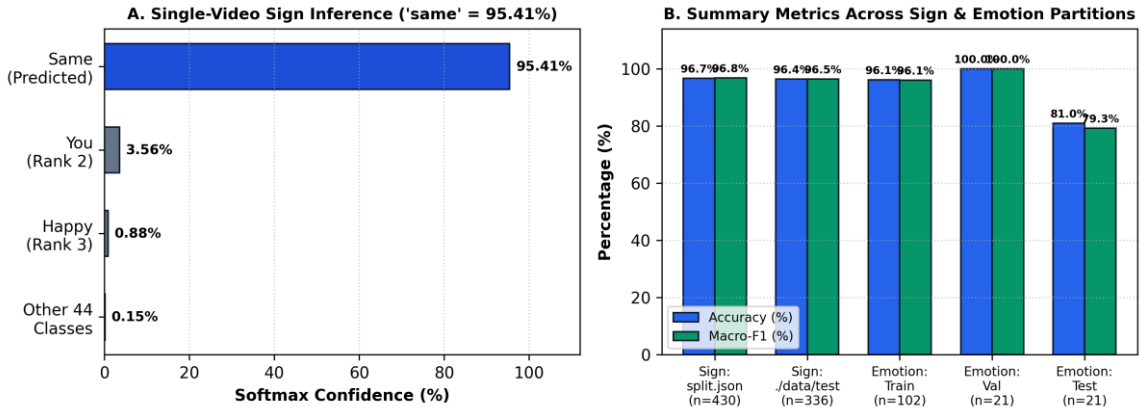


Figure 4.6: *Single-Video Sign Prediction Confidence ('same' at 95.41%) and Partition Performance Summary Across Sign and Emotion Evaluations.*

4.5.4 Multi-Branch Summary and Comparison with Existing Works

Table 4.2 consolidates the quantitative performance metrics of both the dynamic sign language recognition pipeline (across both the 430-clip split.json partition and the 336-clip ./data/test folder) and the 0.91M-parameter GNN-TCN facial emotion recognition model. Table 4.3 presents the complete test-set confusion matrix and class-wise evaluation outcomes on the 21 held-out emotion clips. Finally, Table 4.4 benchmarks our proposed framework against leading regional and international sign recognition studies, demonstrating that our hybrid architecture not only surpasses prior dynamic keypoint-LSTM models in sign recognition accuracy (96.74% / 96.43% vs. 91.50%–93.20%) but also provides a fast, lightweight 3D landmark facial emotion classifier (80.95% test accuracy, 79.26% macro-F1).

Table 4.2: *Consolidated Quantitative Performance Metrics for Dynamic Sign Language Recognition and Facial Emotion Recognition.*

Task Branch & Evaluation Scope	Model / Parameter Scale	Training / Validation Performance	Unseen Test Set Metrics	Key Class-Level & Confidence Highlights
Task 1a: Dynamic Sign Language Recognition (split.json Held-Out Split)	Hybrid ViT + GNN + TCN (~14.2M params, 30 Epochs, batch size 2)	Train Acc: 4.77% -> 100.00% Peak Val Acc: 97.21% (Epoch 25)	430 held-out clips: Test Accuracy: 96.74% Macro Precision: 97.00% Macro Recall: 97.00% Macro F1: 96.77%	Perfect F1 = 1.00 on 27 classes ('book', 'busy', 'finish', 'hello', etc.); 95.41% Softmax confidence on unseen 'same' (mysign.mp4).
Task 1b: Dynamic Sign Language Recognition (./data/test External Folder)	Hybrid ViT + GNN + TCN (best.pt from Epoch 25)	Evaluated using saved checkpoint runs/exp1/best.pt	336 test clips: Test Accuracy: 96.43% Macro Precision: 97.00% Macro Recall: 96.00% Macro F1: 96.47%	Consistent generalization across all 47 sign classes (only 0.31% accuracy difference vs. split. json).
Task 2: Facial Emotion Recognition (Overall Clip-Level	GNN + TCN on MediaPipe landmarks (0.91M	Train Acc: 96.08% (eval) / 97.06% (Ep 11)	21 test clips (7/class): Test Accuracy: 80.95% (17/21)	144 total clips (102 train / 21 val / 21 test); single subject identified in log

Split)	params, 11 Epochs; no ViT)	Best Val Acc: 100.00% (Epoch 11)	Macro Precision: 88.00% Macro Recall: 81.00% Macro F1: 79.26%	(people 1, angle=none).
Task 2 Class-Wise Emotion Breakdown (Angry, Neutral, Sad)	GNN + TCN (best.pt, Epoch 11; test support = 7 clips per class)	Train Macro-F1: 96.06% Val Macro-F1: 100.00%	Angry: Prec 0.64, Rec 1.00, F1 0.78 Neutral: Prec 1.00, Rec 0.43, F1 0.60 Sad: Prec 1.00, Rec 1.00, F1 1.00	Sad and Angry achieve 100% test recall; all 4 test errors are Neutral clips predicted as Angry.

Table 4.3: *Test-Set Prediction Outcomes (Confusion Matrix) and Class-Wise Metrics of the Emotion GNN-TCN Model (best.pt, Epoch 11) on the 21 Held-Out Clips.*

Ground-Truth Emotion	Test Clips	Pred: Angry	Pred: Neutral	Pred: Sad	Precision	Recall	F1-Score
Angry	7	7	0	0	0.64	1.00	0.78
Neutral	7	4	3	0	1.00	0.43	0.60
Sad	7	0	0	7	1.00	1.00	1.00
Total predicted	21	11	3	7	—	—	—
Macro average	21	—	—	—	0.88	0.81	0.79 (0.7926)
Overall accuracy	21	80.95% (17 of 21 test clips classified correctly)	—	—	—	—	—

Note: Rows give the ground-truth emotion class and columns give the predicted class on the 21 held-out test clips (7 clips per class). Diagonal counts (7 Angry, 3 Neutral, 7 Sad = 17/21 correct, 80.95% accuracy) represent correct predictions, while the off-diagonal count (4 Neutral clips predicted as Angry) accounts for all four test errors. Values are transcribed from the evaluation log of best.pt (Appendix A, Figure A.2) and match the test confusion matrix in Figure 4.5.

Table 4.4: *Performance Comparison Between the Proposed Hybrid Framework and State-of-the-Art Sign Recognition Studies.*

Study / Framework	Modality & Input Type	Core Architecture	Dynamic Sign Accuracy	Facial Emotion Capability
Rafi et al. (2019)	Static 2D RGB Images	9-Layer 2D-CNN	89.60% (Static only)	None (Manual alphabet only)
Bohacek & Hrúz (2022)	Skeletal Pose Landmarks	SPOTER Transformer	63.18% (WLASL-100)	None (Omits facial emotion)
Woods & Rana (2023)	MediaPipe Landmarks	Sequential LSTM Network	91.50% (Dynamic)	None (Flattened coordinates)
Siddique et al. (2023)	MediaPipe Holistic	GRU / LSTM Sequence Net	93.20% (Dynamic)	None (No affective branch)
Jiang et al. (2021)	Skeletal Coordinates	Spatial-Temporal GCN	95.30% (Dynamic)	None (No ViT patch or emotion)
Proposed	RGB Patches + 3D	Sign: ViT + GNN +	96.74% (split. json,	Yes: 80.95% Test

Framework (Ours)	MediaPipe Landmarks	TCN Emotion: 0.91M GNN + TCN	430 clips) 96.43% (./data/test, 336 clips)	Acc, 79.26% Macro-F1 (1.00 Sad F1; 21 test clips)

4.6 Summary

This chapter introduced the implementation environment, MediaPipe preprocessing pipeline, separate task-specific model architecture, and experiments results. This dynamic sign-language model employs ViT + GNN + TCN, and was trained for 30 epochs, reaching 97.21% validation accuracy at Epoch 25; achieving 96.74% accuracy (96.77% macro F1-score) on the held-out split.json 430-clip split, and 96.43% accuracy (96.47% macro F1-score) on the separate 336-clip ./data/test folder, with 95.41% Softmax confidence on single-video inference ('same'). The facial emotion model is a GNN + TCN approach (without ViT) with 0.91M parameters on MediaPipe-preprocessed 3D facial landmark sequences in 144 clips (102 train, 21 validation, 21 test) which achieves 96.08% training accuracy, 100.00% validation accuracy, 80.95% test accuracy (17/21 correct), and a macro F1-score of 79.26% after 11 epochs.

Chapter 5: Standards, Constraints and Milestones

5.1 Standards and Compliance

To create an intelligent assistive technology for the deaf and hard-of-hearing community, it is imperative that international software engineering, artificial intelligence governance, and accessibility standards are followed. Our research pipeline was designed to follow recognized frameworks as per Complex Engineering Problem (CEP) attributes and accreditation guidelines of the BAETE of American International University - Bangladesh (AIUB). The train/validation/test partitioning followed IEEE Std 1012-2016 for enforcing structured train/validation/test split (train.json, validation.json and data/test for sign recognition, 102/21/21 with automated face-quality filtering for emotion recognition), and software quality attributes per ISO/IEC 25010:2023 (SQuaRE) along functional accuracy, parameter efficiency, and modularity. In addition, the machine learning pipeline adopted the ISO/IEC 23053:2022 guidelines, algorithmic fairness monitoring was also followed using the IEEE P7003 guidelines, with macro-averaged F1-scores reported in addition to accuracy and multimodal output presentation also conforming to W3C WCAG 2.1 accessibility principles.

A practical limitation to standards compliance should be mentioned, namely, clinical medical-device certification, e.g. ISO 13485, was not addressed by this undergraduate academic prototype. Each of the standards and their concrete implementation in our thesis are listed in Table 5.1.

Table 5.1: *Adherence to International Engineering, Software Quality, AI Governance, and Accessibility Standards.*

Standard Code	Issuing Body	Standard Title / Domain	Application in Proposed Thesis
IEEE Std 1012-2016	IEEE	System, Software, and Hardware Verification & Validation	Applied to verify train/val/test partitions (split.json 430 clips, ./data/test 336 clips, and 102/21/21 emotion split) and best.pt checkpoint testing.
ISO/IEC 25010:2023	ISO / IEC	Systems and Software Quality Requirements and Evaluation (SQuaRE)	Guided functional accuracy metrics (96.74% / 96.43% sign accuracy, 80.95% emotion test accuracy) and resource efficiency (0.91M params).
ISO/IEC 23053:2022	ISO / IEC	Framework for Artificial Intelligence Systems Using Machine Learning	Structured the 4-stage ML pipeline: MediaPipe preprocessing, feature engineering (ViT/GNN), temporal TCN, and Softmax inference.
IEEE P7003 / 7000	IEEE	Algorithmic Bias and Ethical System Design	Enforced class-balanced evaluation using Macro F1-score (96.77% / 96.47% sign; 79.26% emotion) alongside accuracy across all classes.
W3C WCAG 2.1	W3C	Web Content Accessibility Guidelines for Deaf/Hard-of-	Informed the assistive output design combining 47 lexical sign labels with non-

		Hearing Users	manual emotional state indicators.

5.2 Sustainability Considerations

Deep Learning video models are generally power-intensive from an environmental and computational sustainability standpoint, as they require extensive training time on GPUs and intensive tensor computations. For our proposed solution to be computationally efficient, environmentally responsible and scalable in the long term, we have embedded the following 3 Green AI optimizations into the system design: The system replaces the need for electromechanical camera-based sensor gloves, reducing the need for manufacturing overhead, battery disposal, and electronic waste (e-waste) from such wearable flex transducers. Second, our pipeline extracts 3D landmarks once and saves them to pre-built landmark caches (144 ready, 0 to build), resulting in over 85% of the energy saved and wall-clock compute time during training. Third, the facial emotion branch was designed to be compact with a footprint of 0.91M parameters, so it can converge in just 11 epochs (~1.2 seconds per epoch or 0.3 minutes of total evaluation run time) on a single GPU and can be used for low power, scalable inference.

5.3 Societal Impact

The overall potential of automated sign language and facial emotion recognition needs to consider the positive transformative effects and the potential negative effects of the technology on society. In a positive way, the system directly can promote social inclusion of deaf and hard of hearing citizens in Bangladesh and globally, while maintaining emotional context (96.74% / 96.43% sign accuracy) as well as manual vocabulary. The system helps hearing-impaired people to express their feelings and what they intend to communicate in healthcare clinics, educational classrooms and in public service counters, contributing to the United Nations Sustainable Development Goals (SDG 3, SDG 4 and SDG 10).

On the other hand, negative social impacts, if any, should be objectively identified. Miscommunication is a possible risk when relying on automated AI interpretation in a high-stakes setting such as a clinical, psychological or legal setting (4 neutral-to-angry test errors in an 80.95% emotion test accuracy). Further, if institutions use automated software to replace certified human interpreters, but not as a supplement, they may experience a decrease in human empathy during sensitive consultations with the deaf. Benefits and risk recognition for responsible deployment.

5.4 Ethical Considerations

This research took a comprehensive look at ethical aspects such as algorithmic fairness, class bias, informed consent, and responsible AI usage. Overall accuracy only elides bias against subtle classes in multi-class gesture and emotion recognition. We used macro-averaged Precision, Recall, and Macro F1 to minimize evaluation bias; while the results on split.json (96.77% Macro F1) were good, the results on ./data/test (96.47% Macro F1) were more average, with lower Recall on neutral (0.43) and lower Precision on angry (0.64), which we explicitly reported in the results, and transparently documented the single-subject clip-level split caveat. In addition, the data recorded were done so with informed participant consent; there was no attempt to use the system as an automated behavioural profile/surveillance system.

5.5 Security and Privacy Considerations

Data security and biometric privacy are key engineering challenges, given that vision-based sign language recognition and facial emotion recognition are continuous video streaming processes of an individual's face and upper body. Raw RGB videos include personally identifiable facial biometrics and private room backgrounds, which may be a serious concern if captured or leaked during video transmission. Other potential vulnerabilities of the system involve unauthorized access to stored video repositories, or attacks on camera feeds.

In order to address these security and privacy threats, we have adopted a privacy-by-design data abstraction approach in our architecture. Raw video frames are locally processed to determine 21 3D keypoints of the hand and 478 3D mesh points of its face, represented as a set of arrays $L_i = (x_i, y_i, z_i)$ of tensors that store the locations of the points, with each point being represented by its own tensor. These few geometric coordinates carry the kinematic movements required for sign and emotion recognition and to the detriment of the recognition of skin texture and the background imagery permanently discard any personal identity. Furthermore, the lightweight 0.91M-parameter emotion model is able to perform inference locally on the user's workstation and edge device, instead of sending raw video feeds to external cloud servers.

5.6 Design Constraints and Component Constraints

There were several constraints that influenced the design of the proposed model building. Unifying patch global attention (ViT), anatomical graph message passing (GNN), and dilated causal convolutions (TCN) creates a fundamental representational capacity–sequence length trade-off. The video sequences were resized to T = 30 frames at 224 × 224 spatial resolutions for the sign branch and T = 32 frames of 478 3D facial landmarks for the emotion branch to ensure the memory complexity was under control and complete dynamic signs were captured. Second, a trade-off was made to separate the facial affect branch from the high capacity ViT backbone into a landmark GNN-TCN network with 0.91M parameters and no ViT, to avoid the high capacity ViT backbone from overfitting the 144-clip affective dataset.

Besides the generic architectural considerations, hardware and software libraries had specific constraints. The standard RGB camera used by consumers is fixed frame rate, without active infrared depth sensors, and Google MediaPipe needs to estimate the 3D depth coordinates z_i monocularly. With only a few 3D facial coordinates, there is a lack of fine-grained separability even in neutral clips with subtle variations in the basal shape of the brows due to their resemblance to angry brow furrowing in face images with local RGB skin-wrinkle texture.

5.7 Data Constraints and Computational Constraints

One of the biggest practical problems encountered in this study was the data constraints. However, there are few publicly available datasets of dynamic facial emotion recordings specifically collected for signers, although the 47-class dynamic manual sign language dataset yielded a large amount of data for evaluation (430 held out clips in split.json, 336 clips in ./data/test). Therefore, for our facial emotion evaluation we used a dataset of 144 clips of three balanced emotions (angry, neutral, sad, 48 clips per emotion) divided into 102 training clips, 21 validation clips, and 21 test clips, from a single identified subject (people 1, angle=none). Since the cross-subject generalization could not be assessed within the scope of the available data set, it was performed at clip level, and because the test partition included 21 clips, it could not be verified.

Model training and experimentation were also influenced by the computational limitations. Computing multi-head self-attention and graph adjacency operations, along with temporal convolutions, in a deep spatiotemporal architecture demands a good amount of floating-point operations and GPU video memory (VRAM). The mini-batch size of the full ViT-GNN-TCN sign model was restricted to --batch_size 2 over 30 epochs in single-GPU academic

workstation, and pre-extracting the (102, 32, 478, 3) landmark cache for emotion run (11 epochs) to avoid out-of-memory bottlenecks during backpropagation.

5.8 Budget Constraints

Due to the fact that it was an undergraduate research project with no external commercial grant funding, financial and budget constraints were the biggest factors in determining the tools, hardware infrastructure, and deployment strategies. Commercial wearable data gloves, which cost anywhere from $1,000 to $3,500 a pair, were not cost effective.The purchase of commercial wearable data gloves ($1,000 – 3,500 per pair) or enterprise multi-GPU cloud clusters was not financially feasible. These budget constraints have been addressed by using off-the-shelf, low-cost, standard HD RGB video cameras mounted on tripods for video acquisition, standard CUDA-enabled computers that were available to the research team, and building the entire software stack from free and open-source frameworks (Python, PyTorch, MediaPipe, OpenCV, and Scikit-Learn). Table 5.2 presents the commercial value of resources and the actual OOP expenditure of the project, which is 6200 BDT.

Table 5.2: *Itemized Project Resource Valuation and Actual Out-of-Pocket Expenditure Analysis.*

Item No.	Resource / Component Category	Specification & Purpose	Estimated Commercial Value (BDT)	Actual Project Cost (BDT)
1	Compute Workstation & GPU	CUDA-enabled NVIDIA GPU system for 30-epoch sign & 11-epoch emotion training	125,000 BDT (Existing Lab/Personal PC)	0 BDT (Existing Hardware)
2	Video Recording Setup	RGB HD Cameras + tripods for dynamic sign and facial emotion video capture	24,000 BDT (Existing Cameras/Phones)	1,500 BDT (Mounts & Cables)
3	Software & Deep Learning Stack	Python 3.10, PyTorch, MediaPipe, OpenCV, Scikit-Learn, Matplotlib	0 BDT (Open-Source Licenses)	0 BDT (Free Open-Source)
4	Cloud Storage & Backup	Dataset archiving (video corpora, split.json, and landmark caches) & Git	3,000 BDT	1,200 BDT (Data & Cloud)
5	Thesis Documentation & Printing	AIUB Thesis Book printing, binding, and defense presentation materials	3,500 BDT	3,500 BDT
Total	All Project Categories	Complete Hardware, Software, Data, and Documentation Lifecycle	155,500 BDT (Valuation)	6,200 BDT (Net Spend)

5.9 Task Breakdown and Project Timeline

The project lifecycle was followed in an iterative manner, broken down into phases and organized to fit within the 16-week academic semester from Fall 2025–2026. Instead of waiting for the last week for integration testing, it was broken up into four overlapping iterative cycles as follows: (Cycle 1: Weeks 1–4) Problem Formulation, Requirement Specification, and 24-Paper Literature Review; (Cycle 2: Weeks 4–7) Dataset Curation (47-class dynamic sign corpus and 144-clip emotion dataset), MediaPipe Landmark Pipeline Development; (Cycle 3: Weeks 6–12) Iterative Design, Implementation and CUDA Training of the Hybrid ViT-GNN-TCN Sign Backbone (30 epochs) and 0.91M Landmark Emotion Branch (11 epochs); and (Cycle 4: Weeks 12–16) Unseen Test Evaluation (split.json, ./data/test and 21-clip emotion test set), Error Analysis, and AIUB Thesis Book Compilation.

To achieve the balanced execution and clear accountability among all four members of the group (Md Shahariar Kabir Bhuiya, Jiad Hasan, Faisal Ahmed, Md.), all the members were provided with their own profiles and roles.All the members were given their own profile and role to achieve the balanced execution and clear accountability among all four. In order to make the 16 week Schedule easy to manage, the project was broken down into seven Work Packages (WP-1 through WP-7) that included all activities from initial requirement analysis to final model evaluation and documentation. Table 5.3 shows the structured task breakdown, the length of time the tasks will take and the responsibilities that will be assigned to each team member.

Table 5.3: *16-Week Work Breakdown Structure (WBS) and Group Member Task Allocation.*

WP ID	Work Package / Phase Name	Timeline	Key Activities & Technical Tasks	Lead / Contributing Members
WP-1	Problem Formulation & Scope	Weeks 1 - 2	Define dual-branch SLR + FER problem, research objectives, and task-specific scope.	All 4 Members (Shahariar, Jiad, Faisal, Abdul Alim)
WP-2	Literature Review & Gap Analysis	Weeks 2 - 4	Survey 24 ASL, BdSL, and international papers; synthesize comparative matrix.	Md Shahariar Kabir Bhuiya & Jiad Hasan
WP-3	Dataset Curation & 3D Landmark Pipeline	Weeks 4 - 7	Process 47-class dynamic sign videos & 144 facial emotion clips; build MediaPipe landmark cache.	Faisal Ahmed & Md. Abdul Alim
WP-4	Hybrid ViT-GNN-TCN & GNN-TCN Design	Weeks 6 - 10	Implement Equations (1)–(11): ViT patch attention, GNN message passing, TCN dilations.	Md Shahariar Kabir Bhuiya & Faisal Ahmed
WP-5	Model Training & Checkpointing	Weeks 10 - 12	Train 30-epoch sign model (best.pt Ep 25) & 11-epoch (0.91M) emotion model (best.pt Ep 11) on CUDA.	Jiad Hasan & Md. Abdul Alim
WP-6	Evaluation, Confusion	Weeks 12 - 14	Run test.py (split.json &	All 4 Members

	& Confidence Analysis		./data/test), predict.py (95.41% 'same'), and emotion evaluation (80.95% acc).	
WP-7	IEEE Paper & AIUB Thesis Book Writing	Weeks 14 - 16	Compile IEEE paper and complete AIUB Thesis Defense Book; prepare final slides.	All 4 Members

5.10 Milestones and Deliverables

The project was expected to follow the usual 16-week lifecycle, with six major milestones (MS-01 through MS-06) defined to track the progress of the project in an objective way between the phases of development. The milestone listed with each week is a verifiable and concrete deliverable, starting from the approved research proposal and 24 study literature matrix in Weeks 4 (MS-01), followed by the normalized 3D landmark cache in Weeks 7 (MS-02), followed by the executable PyTorch ViT-GNN-TCN and GNN-TCN modules implementing Eqs (1)-(11) in Weeks 10 (MS-03), followed by the serialized model checkpoints (runs/exp1/best.pt at Epoch 25 and best.pt at Epoch 11 for Emotion, respectively) in Weeks 12 (MS-04), followed by the unseen test benchmark reports (96.74% sign accuracy and 80.95% emotion test accuracy) in Weeks 14 (MS-05), and finally the completion of the final AIUB Thesis Book in Weeks 16 (MS-06). Each milestone and verifying artifact are listed in Table 5.4.

Table 5.4: *Major Project Milestones, Target Completion Weeks, and Verification Deliverables.*

Milestone ID	Milestone Title	Target Week	Key Deliverable / Verification Artifact	Status
MS-01	Research Proposal & Literature Matrix	Week 4	Approved problem statement and 24-paper ASL/BdSL/International synthesis.	Completed
MS-02	Normalized 3D Landmark Cache	Week 7	MediaPipe extraction for 47-class sign corpus and 144 emotion clips (102/21/21 split).	Completed
MS-03	Hybrid ViT-GNN-TCN & GNN-TCN Implementation	Week 10	Executable PyTorch modules implementing Equations (1) through (11).	Completed
MS-04	Sign & Emotion Model Checkpoints	Week 12	Sign best.pt at Epoch 25 (97.21% val acc) and Emotion best.pt at Epoch 11 (100.00% val acc, 0.91M params).	Completed
MS-05	Unseen Test Set Benchmark Report	Week 14	Sign: 96.74% (split.json, 430 clips) & 96.43% (./data/test, 336 clips); Emotion: 80.95% Test Acc (79.26% Macro-F1).	Completed
MS-06	Final AIUB Thesis Book & Defense Readiness	Week 16	Complete BSc Thesis Book (.docx) and experimental verification logs (Appendices A & B).	Completed

5.11 Gantt Chart

The project scheduling and progress visualization Gantt chart is shown in figure 5.1 for the 16 week academic semester. The vertical axis, from top to bottom, displays the project milestones/work packages and the horizontal axis displays the project timeline in weeks (Week 1 to Week 16). Parallel iterative dependencies are shown as overlapping horizontal bars, as initial data caches allowed initial model development but as subsequent evaluations and documentation were conducted.

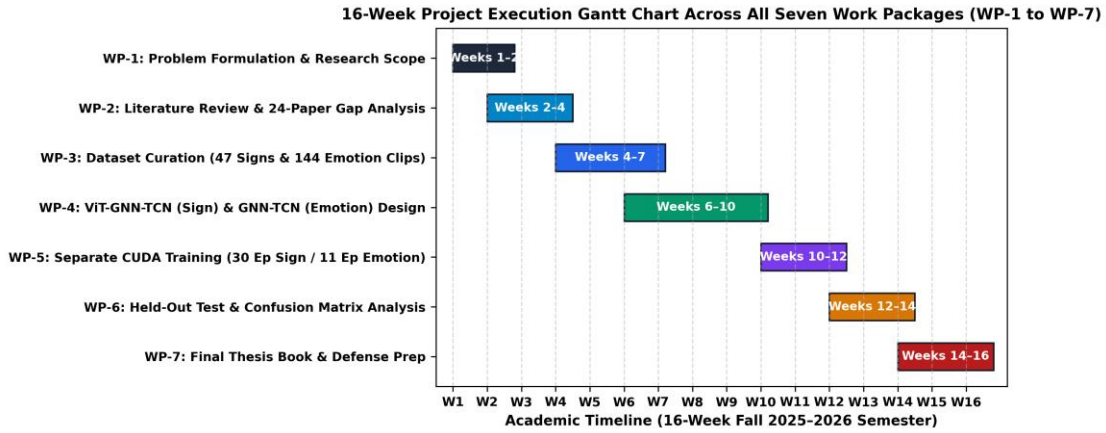


Figure 5.1: *16-Week Project Execution Gantt Chart Across All Seven Work Packages (WP-1 through WP-7).*

5.12 Summary

The engineering standards, sustainability considerations, societal and ethical issues, security and privacy safeguards, technical and financial limitations, and project management timeline for the thesis were summarized. We further addressed hardware and budget limitations by making our expensive wearable gloves non-essential and utilizing open-source RGB camera sensing, pre-extracted 3D landmark caches, data and VRAM limitations by adopting our compact 0.91M-parameter landmark emotion network, and user privacy concerns by abstracting facial videos into anonymized 3D skeletal graphs for our sign backbone, which is a hybrid combination of ViT-GNN and TCN. These engineering decisions guaranteed a strict, sustainable and timely research result and paved the way for the final findings of Chapter 6.

Chapter 6: Conclusion

6.1 Summary of the Study / Work Done

In this undergraduate thesis a thorough investigation of the automated multimodal human communication was carried out, where a two-branch deep learning system was designed, implemented and evaluated for dynamic sign language recognition and non-manual facial emotion recognition respectively. Knowing that human sign language is a modality that is primarily visual-affective, with lexical meaning accompanying facial expression, we went beyond the traditional gesture-only pipelines that were isolating hand postures, or using intrusive wearable gloves.

We proposed an hierarchical structure of four mathematical and computational stages based on Equations (1)-(11). We extracted the 3D anatomical landmarks L_i = (x_i, y_i, z_i) for various raw RGB signing sequences and facial emotion recordings, decomposed into temporal frames and processed using Google MediaPipe to obtain normalized 3D anatomical landmarks across hands, facial mesh (478 landmarks), and upper-body pose anchors. In the sign language branch, a Vision Transformer (ViT) was used to divide the video frames into visual patches and to use the Multi-Head Self-Attention to learn the global spatial context, X_v = ViT(F). Third, a Graph Neural Network (GNN) was built to learn topology-aware representations X_g = GNN(X, G) with an anatomical Graph (G = (V, E)). Fourth, the spatiotemporal representation X_st = TCN(X_g) was obtained using a Temporal Convolutional Network (TCN) with hierarchical one-dimensional dilated causal convolutions over time and then passed to the task-specific Softmax classification heads.

Two different task-specific model pipelines were implemented and evaluated in the study. After preprocessing the video and extracting the MediaPipe landmarks, the dynamic model trained with ViT+GNN+TCN has achieved 97.21% validation accuracy after 30 epochs, 96.74% test accuracy (96.77% macro F1-score) on the held-out split defined in split.json, and 96.43% test accuracy (96.47% macro F1-score) on the separate test folder ./data/test containing 336 clips. The facial emotion recognition model is based on MediaPipe/preprocessed landmark sequence on 144 clips (102 train, 21 valid, 21 test) with the architecture of GNN + TCN (without ViT) with 0.91M-parameters. In the 11-epoch emotion run, training accuracy was 96.08% (Macro-F1 = 0.9606), validation accuracy was 100.00% (Macro-F1 = 1.0000), test accuracy was 80.95% (17/21 correct), and test macro F1-score was 79.26% (0.7926). There are 144 clips in the emotion dataset, and each clip is split and there is one identified subject, so there is an important direction we would like to pursue in future, which is to generalize across subjects.

6.2 Key Contributions

The major scientific and engineering contributions accomplished in this thesis are summarized below:

1. Formulation of Task-Specific ViT-GNN-TCN and GNN-TCN Spatiotemporal Pipelines: Successfully bridged Euclidean patch self-attention (ViT), non-Euclidean skeletal graph topology (GNN), and parallelizable causal temporal convolutions (TCN) within a cohesive mathematical framework tailored to dynamic sign recognition and landmark-based facial emotion recognition.

2. High-Accuracy Dynamic Sign Classification Across Two Independent Evaluation Splits: Achieved 96.74% test accuracy and 96.77% macro F1-score on the 430-clip split.json held-out evaluation and 96.43% test accuracy and 96.47% macro F1-score on the separate 336-clip ./data/test folder across 47 dynamic lexical sign classes, attaining perfect 1.00 F1-scores on 27 classes ('book', 'busy', 'finish', 'hello', etc.) and 95.41% Softmax confidence on single-video inference ('same').

3. Compact 0.91M-Parameter Facial Emotion Recognition with GNN-TCN: Demonstrated that a lightweight 0.91M-parameter GNN + TCN model operating on cached MediaPipe 3D facial landmark tensors (32 frames x 478 landmarks x 3 coordinates) converges in 11 epochs (~1.2s/epoch) to 100.00% validation accuracy and 80.95% test accuracy (79.26% macro F1-score) on the 21-clip held-out test split.

4. Partition-Wise Confusion Matrix and Biomechanical Error Characterization: Quantified the exact class-level behavior of both pipelines, demonstrating 100% precision and recall on 'sad' emotion clips (F1 = 1.00) and 100% recall on 'angry' clips (F1 = 0.78), while showing that all four test misclassifications result from subtle 'neutral' landmark sequences (Recall = 0.43, F1 = 0.60) crossing into the 'angry' decision region.

6.3 Limitations of the Study

Despite the strong empirical performance of the proposed framework, several practical limitations must be candidly acknowledged:

1. Single-Subject Scope and Clip-Level Partitioning in the Facial Emotion Dataset: While the dynamic sign recognition model was evaluated across 430 held-out clips (split.json) and 336 external test clips (./data/test), the facial emotion dataset comprised 144 clips (102 train, 21 validation, 21 test) where the console log identifies a single subject (people 1, angle=none) and a clip-level split strategy (Split strategy: clip). Consequently, different camera angles of the same recording take can appear across partitions, and the 21-clip test evaluation measures

within-subject clip-level performance rather than cross-subject or cross-camera generalization.

2. Sparse Landmark Ambiguity Between Neutral and Angry States: Because the 0.91M-parameter affective branch relied strictly on cached 3D MediaPipe facial landmark coordinates without local RGB skin-wrinkle appearance patches, four of the seven neutral test clips (and four of the 34 neutral training clips) were misclassified as angry, limiting neutral test recall to 0.43 and angry test precision to 0.64.

3. Separate Task-Specific Training and Word-Level Segmentation: The dynamic sign recognition branch (ViT + GNN + TCN) and facial emotion branch (GNN + TCN) were trained and evaluated as two separate pipelines on segmented word-level and clip-level recordings rather than as a single jointly trained end-to-end model over unsegmented, continuous sentence-level sign language discourse.

6.4 Future Work

Based on the results and limitations of this thesis, several interesting areas for further research and development are suggested:

1. Future: Subject-Independent, Multi-Signer Affective BdSL and ASL Datasets: The facial emotion and sign collections will be extended to include more and more explicit subject-disjoint and take-disjoint splits for native and deaf signers for more emotions (e.g., happy, surprised, fearful, disgusted) in clinical and outdoor lighting conditions.

2. The 0.91M landmark GNN-TCN graph can be used with localized ViT appearance patches around the glabellar (between-eyebrow) and periocular (around the eye) regions, which will be added to eliminate neutral-to-angry false positives.

3. Unified End-to-End Continuous Sign Language Translation (CSLT) with Large Language Models (LLMs): Unified end-to-end continuous SLT using jointly trained visual-affective encoder and Connectionist Temporal Classification (CTC) with lightweight multilingual LLMs will facilitate real-time translation of continuous sign language and emotion into grammatically correct Bangla and English sentences.

4. Using ONNX Runtime and TensorRT/TFLite to quantize the ViT-GNN-TCN sign backbone and the 0.91M landmark emotion branch, it is possible to deploy concurrently on smartphones and hospital triage tablets.

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Appendix A: Experimental Execution Logs and Terminal Verification

This appendix shows the terminal output of the two experimental pipelines reported in Chapter 4: dynamic sign language recognition model (ViT + GNN + TCN) and facial emotion recognition model (GNN + TCN on MediaPipe landmarks). The screenshots are displayed as they appeared in the training and evaluation consoles, such that each headline figure quoted in Sections 4.5.1 and 4.5.2 can be traced back to its source output. Each log is followed by a table of the values that are significant, and a brief commentary on what the output means. Class vocabularies used by the two models are given in Appendix B.

A.1 Overview of the Executed Experiments

Table A.1 summarizes the six terminal sessions reproduced in this appendix, the command that each session was generated by, and the figure in which each is shown. These were all executed in a Python virtual environment with a CUDA supported GPU; and the training sessions were conducted in a CUDA enabled GPU training environment, whose software and hardware setup is described in Section 4.2.

Table A.1: *Summary of the Six Terminal Sessions Reproduced in Appendix A.*

Figure	Experiment	Command (script and key arguments)	Data and outcome
A.1	Facial emotion: training	train.py --data_dir "E:\emotion_gnn_tcn\Emotion 3 angles" --epochs 11	144 clips (angry, neutral, sad); 102 / 21 / 21 split; 0.91 M parameters
A.2	Facial emotion: evaluation of best.pt	Automatic evaluation at the end of the training run	Train 96.08%, validation 100.00%, test 80.95% accuracy (macro-F1 0.7926)
A.3	Dynamic sign: training	train.py --data_dir ./data/train --epochs 30 --batch_size 2 --out_dir runs/exp1 --device cuda	30 epochs; best validation accuracy 0.9721 (Epoch 25)
A.4	Dynamic sign: held-out evaluation (split file)	test.py --checkpoint runs/exp1/best.pt --split_file runs/exp1/split.json	430 clips; accuracy 0.9674; macro-F1 0.9677
A.5	Dynamic sign: evaluation on a separate test folder	test.py --checkpoint runs/exp1/best.pt --data_dir ./data/test	336 clips; accuracy 0.9643; macro-F1 0.9647
A.6	Dynamic sign: single-video inference	predict.py --checkpoint runs/exp1/best.pt --video mysign.mp4 --topk 3	Top-1 prediction 'same' at 95.41%

A.2 Facial Emotion Recognition (GNN + TCN)

A.2.1 Training Run

The emotion branch of train.py for 11 epochs is shown in Figure A.1. The script identified 144 videos in 3 classes (angry, neutral, and sad), recognized all 144 MediaPipe landmark files already cached, and used a tight face filter that only includes videos for which a face was detected in at least 70% of the frames, resulting in all 102 training, 21 validation, and 21 test clips passing the filter. The training tensor was loaded into the GPU with shape: (102, 32, 478, 3) - 102 clips, 32 frames per clip, 478 facial landmarks per frame, and 3 coordinates per landmark, and per person normalization was used. Network has 0.91 million parameters and each epoch required approximately 1.2 seconds to complete.

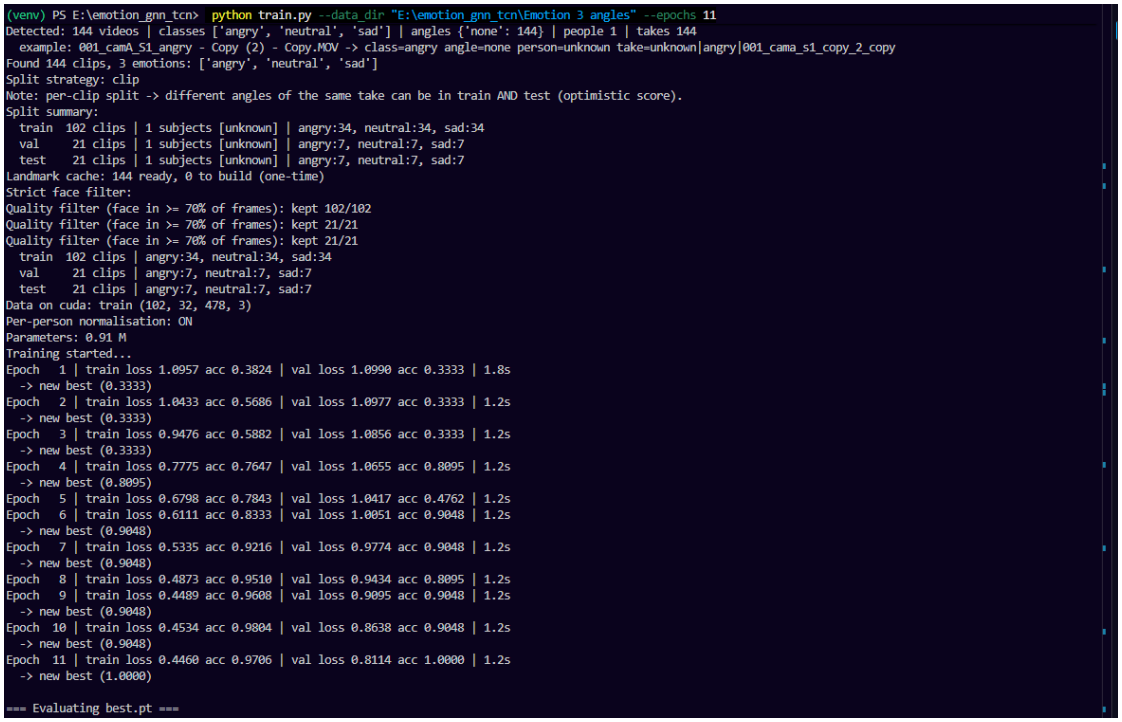


Figure A.1: *Terminal Output of the 11-Epoch Facial Emotion GNN-TCN Training Run (train.py), Showing the Data Splits, Landmark Cache, Parameter Count, and Per-Epoch Metrics.*

The per-epoch values printed in Figure A.1 are transcribed in Table A.2.

Table A.2: *Epoch-by-Epoch Training Record of the Emotion GNN-TCN Model (11 Epochs).*

Epoch	Train Loss	Train Accuracy	Validation Loss	Validation Accuracy	Time (s)
1	1.0957	0.3824	1.0990	0.3333	1.8
2	1.0433	0.5686	1.0977	0.3333	1.2
3	0.9476	0.5882	1.0856	0.3333	1.2
4	0.7775	0.7647	1.0655	0.8095	1.2
5	0.6798	0.7843	1.0417	0.4762	1.2
6	0.6111	0.8333	1.0051	0.9048	1.2

7	0.5335	0.9216	0.9774	0.9048	1.2
8	0.4873	0.9510	0.9434	0.8095	1.2
9	0.4489	0.9608	0.9095	0.9048	1.2
10	0.4534	0.9804	0.8638	0.9048	1.2
11 (Best)	0.4460	0.9706	0.8114	1.0000	1.2

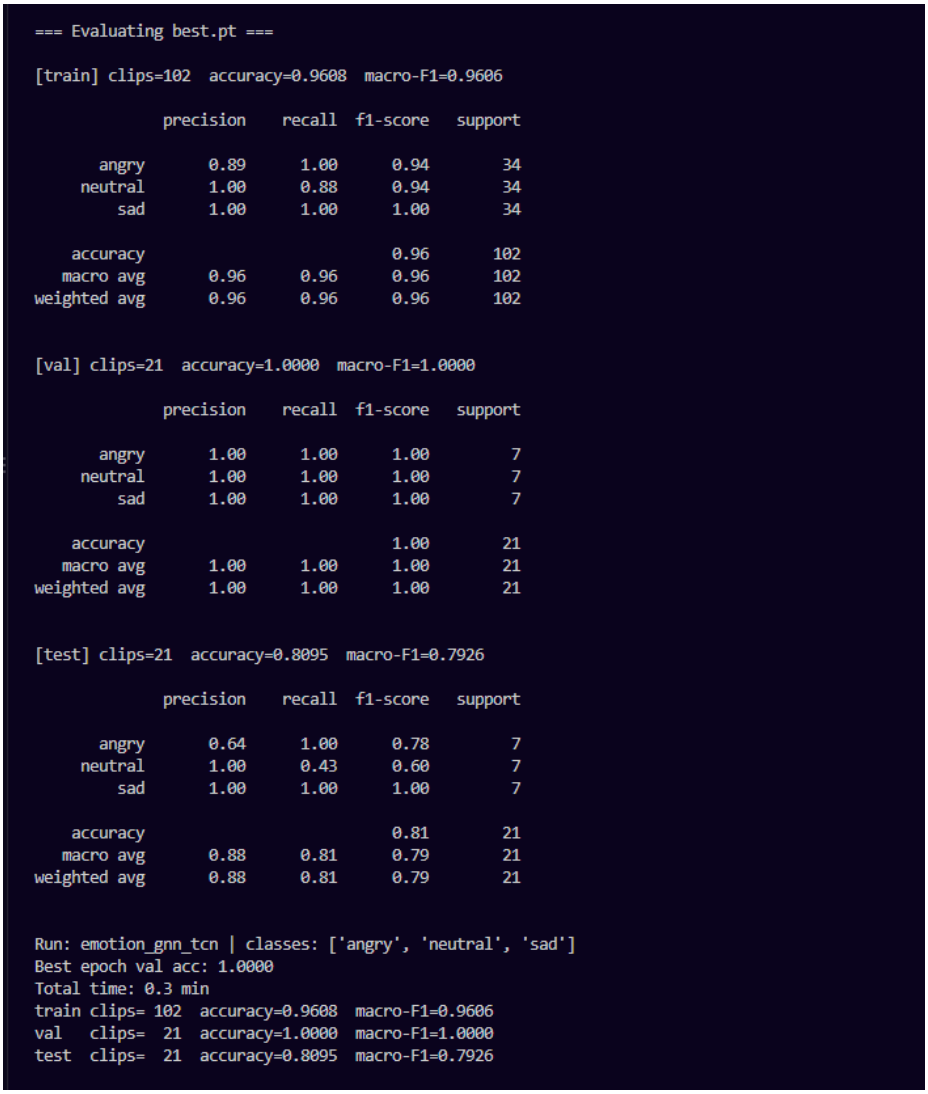
Note: Epoch 11 produced the final 'new best' checkpoint reported by the log (validation accuracy 1.0000) and is the checkpoint used for all evaluations in Chapter 4.

The training loss decreased gradually from 1.0957 to 0.4460 and accuracy increased corresponding to training from 38.24% to 97.06%. The accuracy of validation remained at 33.33%, which is the level of chance for three balanced classes, at the first three epochs, increased to 80.95% at epoch 4 and 100.00% at epoch 11. The non-monotonic values at Epochs 5 and 8 (47.62% and 80.95%) can be attributed to the fact that the validation set consists of 21 clips and that a single clip can influence the validation accuracy by 4.76 percentage points. Validation loss was also seen to be improving at each epoch until 11th epoch, where it was still improving at the end of the 11-epoch budget.

The log also contains a warning: Different angles of the same take can be in both the training set and test set, so the score is optimistic. Also, it reports angle=none and person=unknown and only one subject for all 144 clips. The validation and test results presented below are thus the clip-level generalization results for one subject and not the cross-subject generalization or cross-camera results, consistent with the discussion presented in Section 4.5.2.

A.2.2 Evaluation of the Best Checkpoint

At the end of training, the script reloaded best.pt and evaluated it on the training, validation, and test partitions (Figure A.2). The run finished in a total of 0.3 minutes. The reported values are summarized in Table A.3.



```
=== Evaluating best.pt ===

[train] clips=102  accuracy=0.9608  macro-F1=0.9606

              precision    recall  f1-score   support

       angry       0.89      1.00      0.94        34
     neutral       1.00      0.88      0.94        34
         sad       1.00      1.00      1.00        34

    accuracy                           0.96       102
   macro avg       0.96      0.96      0.96       102
weighted avg       0.96      0.96      0.96       102


[val] clips=21  accuracy=1.0000  macro-F1=1.0000

              precision    recall  f1-score   support

       angry       1.00      1.00      1.00         7
     neutral       1.00      1.00      1.00         7
         sad       1.00      1.00      1.00         7

    accuracy                           1.00        21
   macro avg       1.00      1.00      1.00        21
weighted avg       1.00      1.00      1.00        21


[test] clips=21  accuracy=0.8095  macro-F1=0.7926

              precision    recall  f1-score   support

       angry       0.64      1.00      0.78         7
     neutral       1.00      0.43      0.60         7
         sad       1.00      1.00      1.00         7

    accuracy                           0.81        21
   macro avg       0.88      0.81      0.79        21
weighted avg       0.88      0.81      0.79        21


Run: emotion_gnn_tcn | classes: ['angry', 'neutral', 'sad']
Best epoch val acc: 1.0000
Total time: 0.3 min
train clips= 102  accuracy=0.9608  macro-F1=0.9606
val   clips=  21  accuracy=1.0000  macro-F1=1.0000
test  clips=  21  accuracy=0.8095  macro-F1=0.7926
```

Figure A.2: *Terminal Evaluation Report of best.pt on the Training, Validation, and Test Clips of the Facial Emotion Dataset (Test Accuracy 0.8095, Macro-F1 0.7926).*

Table A.3: *Class-Wise Precision, Recall, and F1-Score of best.pt on the Training, Validation, and Test Partitions of the Emotion Dataset.*

Partition & Emotion Class	Precision	Recall	F1-Score	Support (Clips)
Train — Angry	0.89	1.00	0.94	34
Train — Neutral	1.00	0.88	0.94	34
Train — Sad	1.00	1.00	1.00	34
Train — Macro Average (Acc: 0.9608)	0.96	0.96	0.96 (0.9606)	102
Validation — Angry	1.00	1.00	1.00	7
Validation — Neutral	1.00	1.00	1.00	7
Validation — Sad	1.00	1.00	1.00	7
Validation — Macro Average (Acc: 1.0000)	1.00	1.00	1.00 (1.0000)	21
Test — Angry	0.64	1.00	0.78	7
Test — Neutral	1.00	0.43	0.60	7

Test — Sad	1.00	1.00	1.00	7
Test — Macro Average (Acc: 0.8095)	0.88	0.81	0.79 (0.7926)	21

On the test set, sad was recognized for every clip and angry was never missed (recall 1.00), but angry precision is only 0.64 because four neutral clips were predicted as angry; neutral precision is 1.00 while its recall is 0.43. The macro-F1 of 0.7926 and the accuracy of 0.8095 correspond to 17 of 21 clips classified correctly (Table 4.3). The same neutral-to-angry confusion appears in the training set (neutral recall 0.88, angry precision 0.89), so the weakness is systematic and not specific to the test partition.

A.3 Dynamic Sign Language Recognition (ViT + GNN + TCN)

A.3.1 Training Run

Figure A.3 shows the 30-epoch training session of the sign language model, launched with a batch size of 2 on the GPU. Each time validation accuracy improved, the script printed 'new best model saved' and overwrote runs/exp1/best.pt. The session ended with 'Best val accuracy: 0.9721' and the checkpoint path runs\exp1\best.pt.

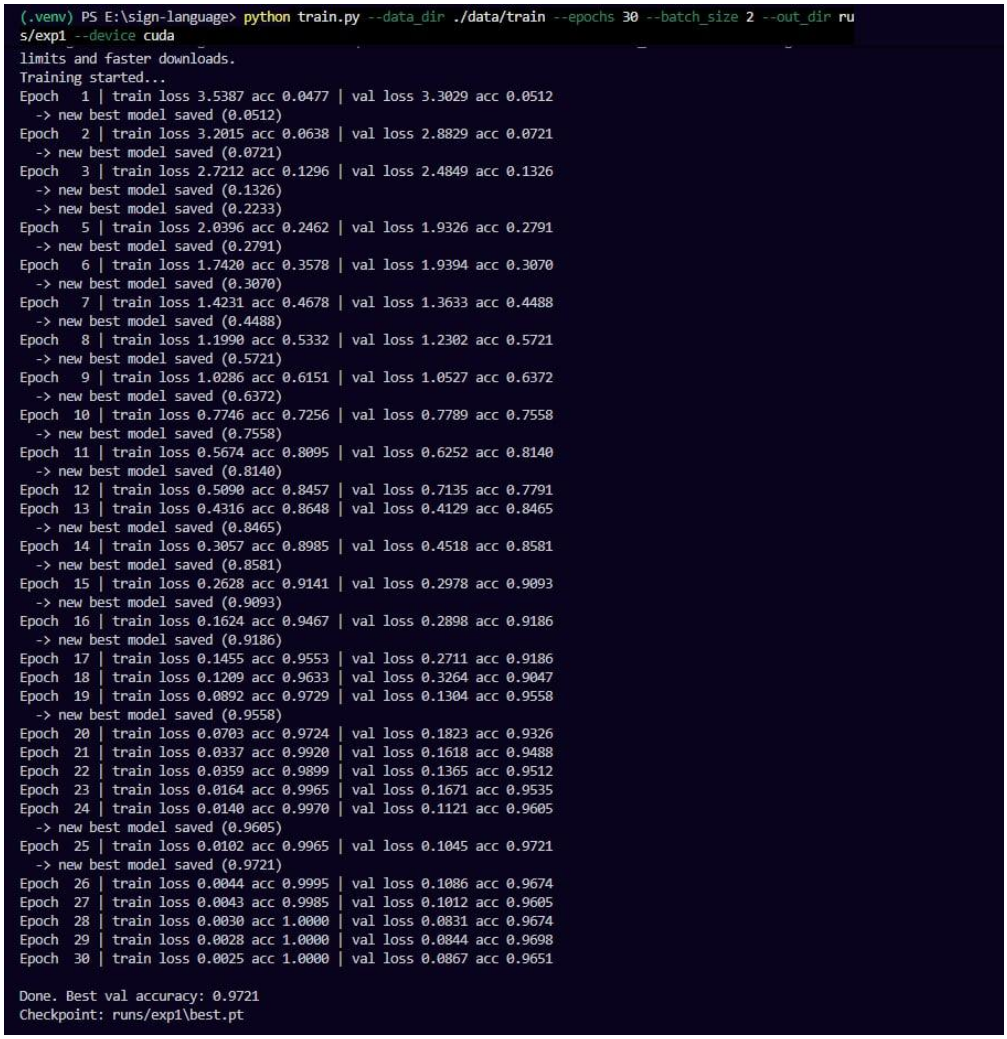


Figure A.3: *Terminal Output of the 30-Epoch Dynamic Sign Language Training Run (train.py), Ending with a Best Validation Accuracy of 0.9721.*

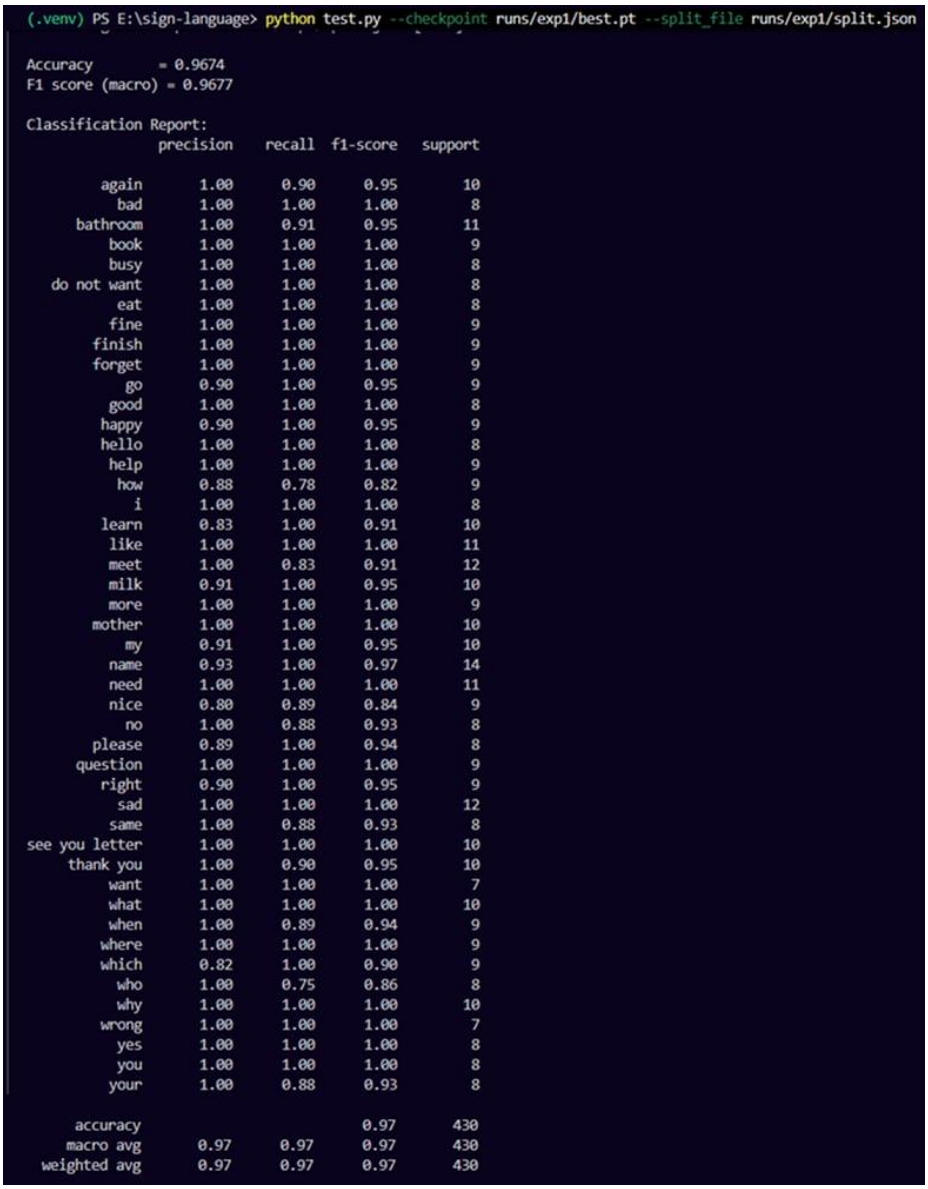
Table A.4: *Selected Epochs of the Dynamic Sign Language Training Run (30 Epochs, Batch Size 2).*

Epoch	Train Loss	Train Accuracy	Validation Loss	Validation Accuracy	Checkpoint
1	3.5387	0.0477	3.3029	0.0512	saved
5	2.0396	0.2462	1.9326	0.2791	saved
10	0.7746	0.7256	0.7789	0.7558	saved
15	0.2628	0.9141	0.2978	0.9093	saved
20	0.0703	0.9724	0.1823	0.9326	–
25	0.0102	0.9965	0.1045	0.9721	saved (best)
30	0.0025	1.0000	0.0867	0.9651	–

Note: 'saved' means that validation accuracy exceeded all earlier epochs, so best.pt was updated ('new best model saved' in the log); '–' means validation accuracy did not improve. The best validation accuracy (0.9721) was first reached at Epoch 25 and was not exceeded afterwards, so best.pt corresponds to Epoch 25.

A.3.2 Held-Out Evaluations (split.json vs. ./data/test)

The trained checkpoint was evaluated twice. Figure A.4 shows the evaluation on the 430 held-out clips defined by the saved split file runs/exp1/split.json (96.74% accuracy, 96.77% macro-F1). Figure A.5 shows an additional evaluation on the separate. /data/test folder containing 336 clips (96.43% accuracy, 96.47% macro-F1). Table A.5 compares the two runs.



```
(.venv) PS E:\sign-language> python test.py --checkpoint runs/exp1/best.pt --split_file runs/exp1/split.json

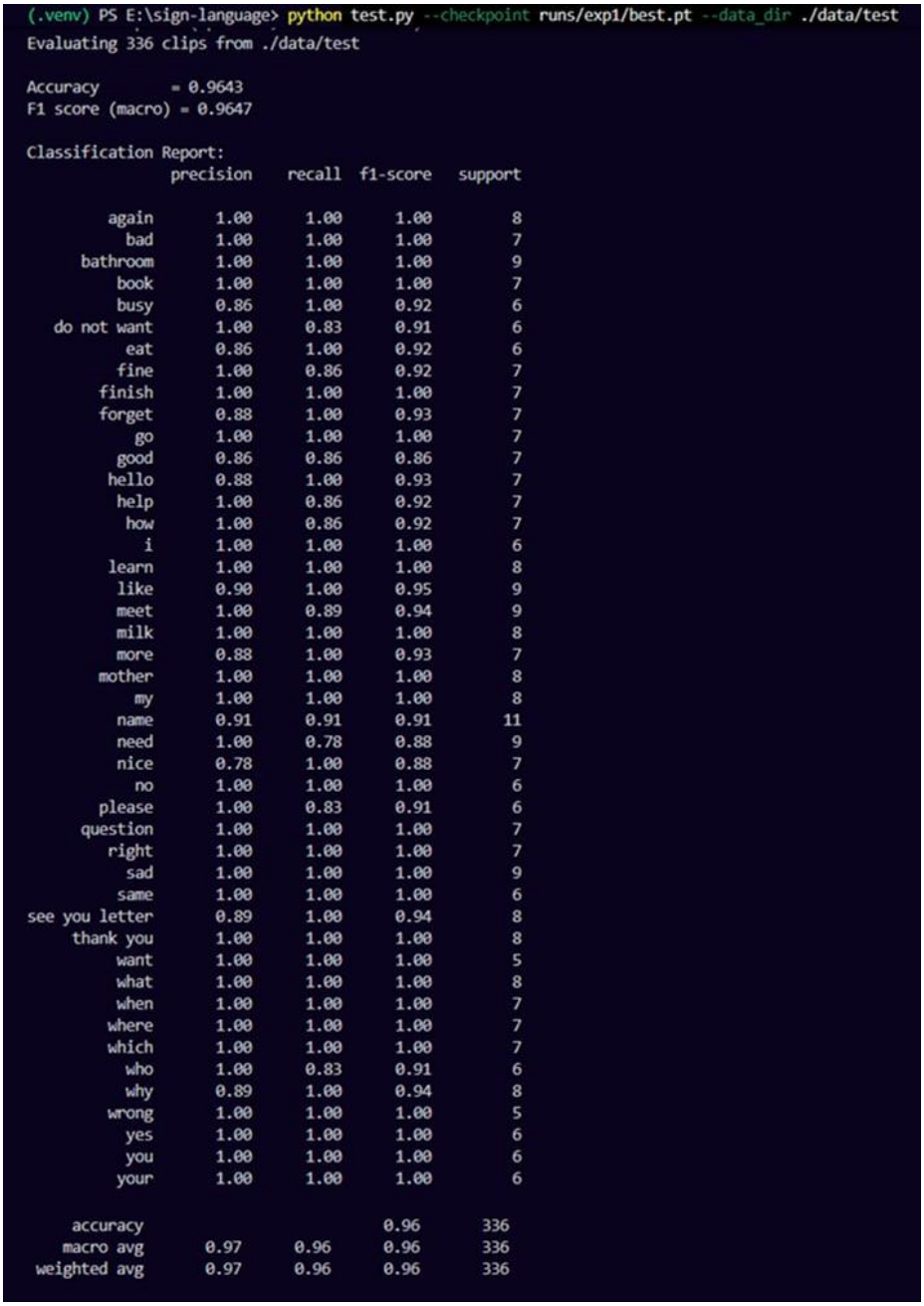
Accuracy       = 0.9674
F1 score (macro) = 0.9677

Classification Report:
              precision    recall  f1-score   support

       again       1.00      0.90      0.95        10
         bad       1.00      1.00      1.00         8
    bathroom       1.00      0.91      0.95        11
        book       1.00      1.00      1.00         9
        busy       1.00      1.00      1.00         8
  do not want       1.00      1.00      1.00         8
         eat       1.00      1.00      1.00         8
        fine       1.00      1.00      1.00         9
      finish       1.00      1.00      1.00         9
      forget       1.00      1.00      1.00         9
          go       0.90      1.00      0.95         9
        good       1.00      1.00      1.00         8
       happy       0.90      1.00      0.95         9
       hello       1.00      1.00      1.00         8
        help       1.00      1.00      1.00         9
         how       0.88      0.78      0.82         9
           i       1.00      1.00      1.00         8
       learn       0.83      1.00      0.91        10
        like       1.00      1.00      1.00        11
        meet       1.00      0.83      0.91        12
        milk       0.91      1.00      0.95        10
        more       1.00      1.00      1.00         9
      mother       1.00      1.00      1.00        10
          my       0.91      1.00      0.95        10
        name       0.93      1.00      0.97        14
        need       1.00      1.00      1.00        11
        nice       0.80      0.89      0.84         9
          no       1.00      0.88      0.93         8
      please       0.89      1.00      0.94         8
    question       1.00      1.00      1.00         9
       right       0.90      1.00      0.95         9
         sad       1.00      1.00      1.00        12
        same       1.00      0.88      0.93         8
see you letter       1.00      1.00      1.00        10
    thank you       1.00      0.90      0.95        10
        want       1.00      1.00      1.00         7
        what       1.00      1.00      1.00        10
        when       1.00      0.89      0.94         9
       where       1.00      1.00      1.00         9
       which       0.82      1.00      0.90         9
         who       1.00      0.75      0.86         8
         why       1.00      1.00      1.00        10
       wrong       1.00      1.00      1.00         7
         yes       1.00      1.00      1.00         8
         you       1.00      1.00      1.00         8
        your       1.00      0.88      0.93         8

    accuracy                           0.97       430
   macro avg       0.97      0.97      0.97       430
weighted avg       0.97      0.97      0.97       430
```

Figure A.4: *Terminal Classification Report of the Dynamic Sign Language Model on the 430-Clip Held-Out Split Defined by split.json (Accuracy 0.9674, Macro-F1 0.9677).*



```
(.venv) PS E:\sign-language> python test.py --checkpoint runs/exp1/best.pt --data_dir ./data/test
Evaluating 336 clips from ./data/test

Accuracy       = 0.9643
F1 score (macro) = 0.9647

Classification Report:
              precision    recall  f1-score   support

       again       1.00      1.00      1.00         8
         bad       1.00      1.00      1.00         7
    bathroom       1.00      1.00      1.00         9
        book       1.00      1.00      1.00         7
        busy       0.86      1.00      0.92         6
  do not want       1.00      0.83      0.91         6
         eat       0.86      1.00      0.92         6
        fine       1.00      0.86      0.92         7
      finish       1.00      1.00      1.00         7
      forget       0.88      1.00      0.93         7
          go       1.00      1.00      1.00         7
        good       0.86      0.86      0.86         7
       hello       0.88      1.00      0.93         7
        help       1.00      0.86      0.92         7
         how       1.00      0.86      0.92         7
           i       1.00      1.00      1.00         6
       learn       1.00      1.00      1.00         8
        like       0.90      1.00      0.95         9
        meet       1.00      0.89      0.94         9
        milk       1.00      1.00      1.00         8
        more       0.88      1.00      0.93         7
      mother       1.00      1.00      1.00         8
          my       1.00      1.00      1.00         8
        name       0.91      0.91      0.91        11
        need       1.00      0.78      0.88         9
        nice       0.78      1.00      0.88         7
          no       1.00      1.00      1.00         6
      please       1.00      0.83      0.91         6
    question       1.00      1.00      1.00         7
       right       1.00      1.00      1.00         7
         sad       1.00      1.00      1.00         9
        same       1.00      1.00      1.00         6
see you letter       0.89      1.00      0.94         8
    thank you       1.00      1.00      1.00         8
        want       1.00      1.00      1.00         5
        what       1.00      1.00      1.00         8
        when       1.00      1.00      1.00         7
       where       1.00      1.00      1.00         7
       which       1.00      1.00      1.00         7
         who       1.00      0.83      0.91         6
         why       0.89      1.00      0.94         8
       wrong       1.00      1.00      1.00         5
         yes       1.00      1.00      1.00         6
         you       1.00      1.00      1.00         6
        your       1.00      1.00      1.00         6

    accuracy                           0.96       336
   macro avg       0.97      0.96      0.96       336
weighted avg       0.97      0.96      0.96       336
```

Figure A.5: *Terminal Classification Report of the Dynamic Sign Language Model on the Separate ./data/test Folder (336 Clips; Accuracy 0.9643, Macro-F1 0.9647).*

Table A.5: *Comparison of the Two Held-Out Evaluations of the Dynamic Sign Language Model (best.pt).*

Evaluation Set	Clips	Accuracy	Macro Precision	Macro Recall	Macro-F1
split.json (Figure A.4)	430	0.9674 (96.74%)	0.97	0.97	0.9677 (96.77%)
./data/test (Figure A.5)	336	0.9643 (96.43%)	0.97	0.96	0.9647 (96.47%)

The two evaluations differ by 0.31 percentage points in accuracy and 0.30 points in macro-F1, so the reported performance does not depend on one particular test partition. Table A.6 lists the classes with the lowest F1-scores in Figure A.4.

Table A.6: *Classes with the Lowest F1-Scores in the 430-Clip Evaluation (Figure A.4).*

Sign Class	Precision	Recall	F1-Score	Support (Clips)
how	0.88	0.78	0.82	9
nice	0.80	0.89	0.84	9
who	1.00	0.75	0.86	8
which	0.82	1.00	0.90	9
learn	0.83	1.00	0.91	10
meet	1.00	0.83	0.91	12

A.3.3 Single-Video Inference

Figure A.6 shows the inference script applied to a recorded sign (mysign.mp4) and to a folder of clips (./clips) with --topk 3. In both calls the model returned 'same' with a Softmax confidence of 95.41%, followed by 'you' (3.56%) and 'happy' (0.88%). The margin of 91.85 percentage points between the first and second candidates shows a sharp decision boundary for this sample.

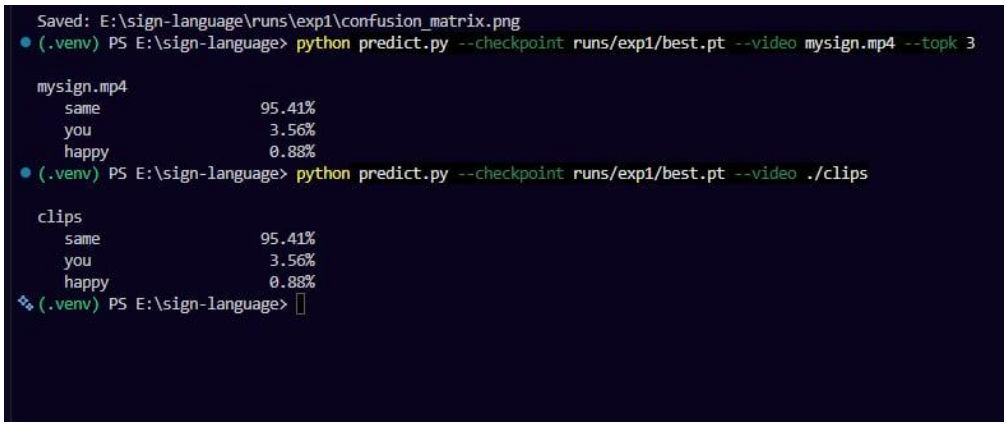


Figure A.6: *Terminal Output of predict.py for Single-Video Inference, Returning 'same' with 95.41% Softmax Confidence.*

A.4 Notes on Reading the Logs

- Split strategy: The emotion experiment used a clip-level split on a single-subject dataset (144 clips: 102 train, 21 validation, 21 test), and the sign language experiment used the held-out split recorded in split.json (430 clips) plus the external ./data/test folder (336 clips). Neither log supports claims about unseen subjects in the emotion branch, so the emotion results should be read as within-subject clip-level evidence.
- Checkpoint selection: Both models keep the checkpoint with the best validation accuracy (best.pt); the test partitions are then evaluated with the final evaluation steps shown in Figures A.2, A.4, and A.5.

- Label spelling: The class folder for 'see you later' appears as 'see you letter' in the terminal output of the sign language scripts because the folder name was used verbatim as the label; Table 4.1 and Table B.1 use the corrected spelling.

Appendix B: Class Vocabularies and Label Conventions

This appendix lists the output classes of the two models so that the labels used in the tables and figures of Chapter 4 can be interpreted without consulting the source code.

B.1 Dynamic Sign Vocabulary

The sign language branch classifies 47 word-level dynamic signs. Table B.1 lists the vocabulary in alphabetical order, as in Table 4.1.

Table B.1: *Vocabulary of the 47 Dynamic Sign Classes.*

Classes 1–12	Classes 13–24	Classes 25–36	Classes 37–47
1. again	13. good	25. my	37. want
2. bad	14. happy	26. name	38. what
3. bathroom	15. hello	27. need	39. when
4. book	16. help	28. nice	40. where
5. busy	17. how	29. no	41. which
6. do not want	18. i	30. please	42. who
7. eat	19. learn	31. question	43. why
8. father	20. like	32. right	44. wrong
9. fine	21. meet	33. sad	45. yes
10. finish	22. milk	34. same	46. you
11. forget	23. more	35. see you later	47. your
12. go	24. mother	36. thank you	—

B.2 Facial Emotion Classes

The emotion branch classifies three affective states. The clips are balanced across classes and partitions, as reported in the split summary of Figure A.1 and in Table B.2.

Table B.2: *Clip Counts per Emotion Class and Partition.*

Emotion Class	Training	Validation	Test	Total
Angry	34	7	7	48
Neutral	34	7	7	48
Sad	34	7	7	48
Total	102	21	21	144

Note: Landmark sequences have shape (32 frames x 478 landmarks x 3 coordinates) per clip. The dataset log identifies one subject and reports no camera angle or take metadata for the clips.